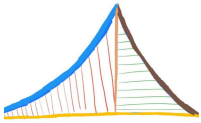
Math for EE



G. V. V. Sharma

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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First manual appeared in January 2018
First edition published on July 10, 2024

In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

CONTENTS

1 CBSE Mathematics 2017 4
1.1 Class 10 4
1.1.1 Formulae 4
1.1.2 Problems 6
1.2 Class 12 22
1.2.1 Formulae 22
1.2.2 Problems 27

1. CBSE MATHEMATICS 2017

A. Class 10

1) Formulae:

1.1.1 A tangent to a circle with centre $\mathbf{O}$ is perpendicular to the radius through the point of contact $\mathbf{T}$,

$$(\mathbf{P}-\mathbf{T})^{\top}(\mathbf{T}-\mathbf{O})=0 \tag{1.1.1.1}$$

where $\mathbf{P}$ is any other point on the tangent.

1.1.2 The lengths of the two tangents drawn from an external point $\mathbf{P}$ to a circle of radius a and centre $\mathbf{O}$ are equal and

$$\|\mathbf{P}-\mathbf{T}_1\|=\|\mathbf{P}-\mathbf{T}_2\|=\sqrt{\|\mathbf{P}-\mathbf{O}\|^2-a^2} \tag{1.1.2.1}$$

where $\mathbf{T}_1$ and $\mathbf{T}_2$ are the points of contact.

Solution: From (1.1.1.1) and Pythagoras' theorem in the right triangle $\mathbf{OT}_i\mathbf{P}$, $i=1,2$,

$$\|\mathbf{P}-\mathbf{O}\|^2=\|\mathbf{P}-\mathbf{T}_i\|^2+\|\mathbf{T}_i-\mathbf{O}\|^2=\|\mathbf{P}-\mathbf{T}_i\|^2+a^2 \tag{1.1.2.2}$$

which is the same for $i=1,2$, yielding (1.1.2.1).

1.1.3 In a right triangle with an acute angle θ,

$$\sin\theta=\frac{\text{opposite}}{\text{hypotenuse}},\quad \tan\theta=\frac{\text{opposite}}{\text{adjacent}} \tag{1.1.3.1}$$

1.1.4 For complementary angles,

$$\tan\left(90^\circ-\theta\right)=\frac{1}{\tan\theta} \tag{1.1.4.1}$$

1.1.5 The locus of a point $\mathbf{x}$ equidistant from the points $\mathbf{A}$ and $\mathbf{B}$ is

$$(\mathbf{A}-\mathbf{B})^{\top}\mathbf{x}=\frac{\|\mathbf{A}\|^2-\|\mathbf{B}\|^2}{2} \tag{1.1.5.1}$$

Solution:

$$\|\mathbf{x}-\mathbf{A}\|^2=\|\mathbf{x}-\mathbf{B}\|^2 \tag{1.1.5.2}$$

$$\implies \|\mathbf{x}\|^2-2\mathbf{A}^{\top}\mathbf{x}+\|\mathbf{A}\|^2=\|\mathbf{x}\|^2-2\mathbf{B}^{\top}\mathbf{x}+\|\mathbf{B}\|^2 \tag{1.1.5.3}$$

which simplifies to (1.1.5.1).

1.1.6 For the quadratic equation $px^2+qx+r=0$ with roots α and β,

$$\alpha+\beta=-\frac{q}{p},\quad \alpha\beta=\frac{r}{p} \tag{1.1.6.1}$$

1.1.7 The nth term of an A.P. with first term a and common difference d is

$$a_n=a+(n-1)\,d \tag{1.1.7.1}$$

1.1.8 The area of a circle of radius r and of a semicircle of radius r are

$$A_{\text{circle}} = \pi r^2, \quad A_{\text{semicircle}} = \frac{1}{2}\pi r^2 \tag{1.1.8.1}$$

1.1.9 The point $\mathbf{R}$ dividing the line segment joining $\mathbf{P}$ and $\mathbf{Q}$ in the ratio $k : 1$ is

$$\mathbf{R} = \frac{k\mathbf{Q} + \mathbf{P}}{k+1} \tag{1.1.9.1}$$

1.1.10 The points $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are collinear if and only if

$$\text{rank}\begin{pmatrix}\mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A}\end{pmatrix} = 1 \tag{1.1.10.1}$$

which, for points in a plane, is equivalent to

$$\left|\begin{pmatrix}\mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A}\end{pmatrix}\right| = 0 \tag{1.1.10.2}$$

1.1.11 The roots of the quadratic equation $px^2 + qx + r = 0$ are

$$x = \frac{-q \pm \sqrt{q^2 - 4pr}}{2p} \tag{1.1.11.1}$$

The roots are equal if and only if

$$q^2 - 4pr = 0 \tag{1.1.11.2}$$

1.1.12 The sum of the first n terms of an A.P. with first term a and common difference d is

$$S_n = \frac{n}{2}\left[2a + (n-1)\,d\right] \tag{1.1.12.1}$$

1.1.13 The volume of a cuboid of length l, breadth b and height h is lbh. Water flowing through a canal of cross-sectional area A with speed v for a time t has volume

$$V = A\,v\,t \tag{1.1.13.1}$$

1.1.14 The volume of a hollow cylinder of outer radius R, inner radius r and length h is

$$V = \pi\left(R^2 - r^2\right)h \tag{1.1.14.1}$$

1.1.15 The area of a sector of angle θ of a circle of radius r and the area of a ring between circles of radii R and r are

$$A_{\text{sector}} = \frac{\theta}{360^\circ}\pi r^2, \quad A_{\text{ring}} = \pi\left(R^2 - r^2\right) \tag{1.1.15.1}$$

1.1.16 For a cone of radius r and height h, the slant height l and the curved surface area are

$$l = \sqrt{h^2 + r^2}, \quad A_{\text{cone}} = \pi r l \tag{1.1.16.1}$$

The curved surface area of a hemisphere of radius r is

$$A_{\text{hemisphere}} = 2\pi r^2 \tag{1.1.16.2}$$

1.1.17 For an event E with m favourable outcomes out of N equally likely outcomes, and

its complement E',

$$\Pr(E) = \frac{m}{N}, \quad \Pr(E) + \Pr(E') = 1 \tag{1.1.17.1}$$

1.1.18 The volume of a cylinder of radius r and height h is

$$V = \pi r^2 h \tag{1.1.18.1}$$

1.1.19 The mth term of an A.P. is related to the sum of its first $2m-1$ terms by

$$S_{2m-1} = (2m-1)\, a_m \tag{1.1.19.1}$$

Solution: From (1.1.12.1), with $n = 2m-1$,

$$S_{2m-1} = \frac{2m-1}{2}\left[2a + (2m-2)\, d\right] = (2m-1)\left(a + (m-1)\, d\right) \tag{1.1.19.2}$$

yielding (1.1.19.1) from (1.1.7.1).

1.1.20 If a person completes a work in t days, the rate of work is $\frac{1}{t}$ of the work per day. The rates of persons working together add up,

$$\frac{1}{t} = \frac{1}{t_1} + \frac{1}{t_2} \tag{1.1.20.1}$$

1.1.21 The angle of depression of an object from a point at a height h is equal to the angle of elevation of that point from the object (alternate angles). If the horizontal distance is d, from (1.1.3.1),

$$\tan\theta = \frac{h}{d} \tag{1.1.21.1}$$

1.1.22 An angle in a semicircle is a right angle. If BC is a diameter of a circle and A is a point on the circle, then

$$\angle BAC = 90^\circ, \quad BC^2 = AB^2 + AC^2 \tag{1.1.22.1}$$

The area of a right triangle with perpendicular sides p and q is $\frac{1}{2}pq$.

2) Problems:

1.1.1 If the angle between two tangents drawn from an external point P to a circle of radius a and centre O, is 60°, then find the length of OP. (CBSE 2017)

Solution: Let $\mathbf{T}_1$ and $\mathbf{T}_2$ be the points of contact of the tangents from $\mathbf{P}$. From (1.1.2.1),

$$\|\mathbf{P} - \mathbf{T}_1\| = \|\mathbf{P} - \mathbf{T}_2\| \tag{1.1.1.1}$$

Also $\|\mathbf{O} - \mathbf{T}_1\| = \|\mathbf{O} - \mathbf{T}_2\| = a$ and $\mathbf{OP}$ is common. So the triangles $\mathbf{OT}_1\mathbf{P}$ and $\mathbf{OT}_2\mathbf{P}$ are congruent, and $\mathbf{OP}$ bisects the angle between the tangents,

$$\angle \mathbf{OPT}_1 = \frac{60^\circ}{2} = 30^\circ \tag{1.1.1.2}$$

From (1.1.1.1), the triangle $\mathbf{OT_1P}$ is right angled at $\mathbf{T}_1$. Using (1.1.3.1),

$$\sin 30^\circ = \frac{a}{\|\mathbf{O}-\mathbf{P}\|} \tag{1.1.1.3}$$

$$\implies \|\mathbf{O}-\mathbf{P}\| = \frac{a}{\frac{1}{2}} = 2a \tag{1.1.1.4}$$

1.1.2 A circle touches all the four sides of a quadrilateral ABCD. Prove that

$$AB + CD = BC + DA$$

(CBSE 2017)

Solution: Let the circle touch AB, BC, CD and DA at $\mathbf{P}$, $\mathbf{Q}$, $\mathbf{R}$ and $\mathbf{S}$ respectively. The tangents drawn from each of $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{D}$ are equal, from (1.1.2.1). Hence,

$$AP = AS, \quad BP = BQ, \quad CR = CQ, \quad DR = DS \tag{1.1.2.1}$$

Adding the above,

$$(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ) \tag{1.1.2.2}$$

Since $\mathbf{P}$ lies on AB, $\mathbf{R}$ on CD, $\mathbf{S}$ on DA and $\mathbf{Q}$ on BC,

$$AP + BP = AB, \quad CR + DR = CD, \quad AS + DS = DA, \quad BQ + CQ = BC \tag{1.1.2.3}$$

Substituting in the above,

$$AB + CD = BC + DA \tag{1.1.2.4}$$

1.1.3 If the distances of $\mathbf{P}\,(x, y)$ from $\mathbf{A}\,(5, 1)$ and $\mathbf{B}\,(-1, 5)$ are equal, then prove that $3x = 2y$. (CBSE 2017)

Solution: Let

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} -1 \\ 5 \end{pmatrix} \tag{1.1.3.1}$$

Then

$$\mathbf{A} - \mathbf{B} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}, \quad \|\mathbf{A}\|^2 = 26, \quad \|\mathbf{B}\|^2 = 26 \tag{1.1.3.2}$$

Substituting in (1.1.5.1),

$$\begin{pmatrix} 6 & -4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \frac{26 - 26}{2} = 0 \tag{1.1.3.3}$$

$$\implies 6x - 4y = 0 \tag{1.1.3.4}$$

$$\therefore 3x = 2y \tag{1.1.3.5}$$

1.1.4 Find the value of p, for which one root of the quadratic equation $px^2 - 14x + 8 = 0$ is 6 times the other. (CBSE 2017)

Solution: Let the roots be α and 6α. From (1.1.6.1),

$$\alpha + 6\alpha = \frac{14}{p} \tag{1.1.4.1}$$

$$\alpha\,(6\alpha) = \frac{8}{p} \tag{1.1.4.2}$$

From (1.1.4.1),

$$\alpha = \frac{2}{p} \tag{1.1.4.3}$$

Substituting in (1.1.4.2),

$$\frac{24}{p^2} = \frac{8}{p} \tag{1.1.4.4}$$

$$\implies p = 3 \tag{1.1.4.5}$$

For $p = 3$, the roots are 4 and $\frac{2}{3}$, and indeed $4 = 6\left(\frac{2}{3}\right)$. The following code verifies the solution

codes/quadratic.c

1.1.5 For what value of n, are the nth terms of two A.Ps $63, 65, 67, \ldots$ and $3, 10, 17, \ldots$ equal ? (CBSE 2017)

Solution: For the first A.P., $a = 63$, $d = 2$ and for the second A.P., $a = 3$, $d = 7$. Equating the nth terms using (1.1.7.1),

$$63 + (n-1)\,2 = 3 + (n-1)\,7 \tag{1.1.5.1}$$

$$\implies 5\,(n-1) = 60 \tag{1.1.5.2}$$

$$\therefore n = 13 \tag{1.1.5.3}$$

Both terms are then equal to 87. The following code verifies the solution

codes/ap_equal.c

1.1.6 On a straight line passing through the foot of a tower, two points C and D are at distances of 4 m and 16 m from the foot respectively. If the angles of elevation from C and D of the top of the tower are complementary, then find the height of the tower. (CBSE 2017)

Solution: Let h be the height of the tower and θ be the angle of elevation from $\mathbf{C}$. The angle of elevation from $\mathbf{D}$ is then $90^\circ - \theta$. From (1.1.3.1),

$$\tan\theta = \frac{h}{4}, \quad \tan\left(90^\circ - \theta\right) = \frac{h}{16} \tag{1.1.6.1}$$

From (1.1.4.1),

$$\frac{h}{4} \cdot \frac{h}{16} = 1 \tag{1.1.6.2}$$

$$\implies h^2 = 64 \tag{1.1.6.3}$$

$$\therefore h = 8 \text{ m} \tag{1.1.6.4}$$

The following code generates Fig. 1.1.6.1

codes/trig_height.py

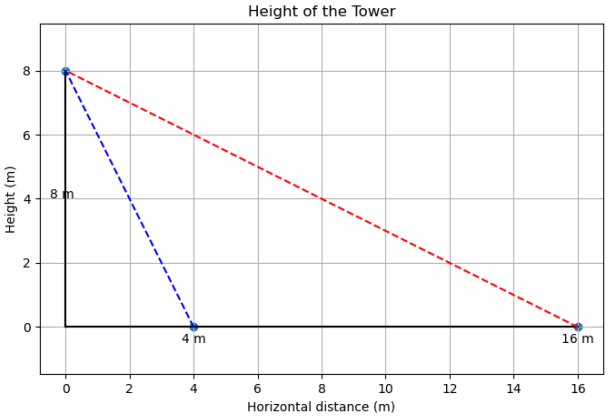


Fig. 1.1.6.1: Tower and the angles of elevation

1.1.7 Three semicircles each of diameter 3 cm, a circle of diameter 4.5 cm and a semicircle of radius 4.5 cm are drawn in the given figure. Find the area of the shaded region. (CBSE 2017)

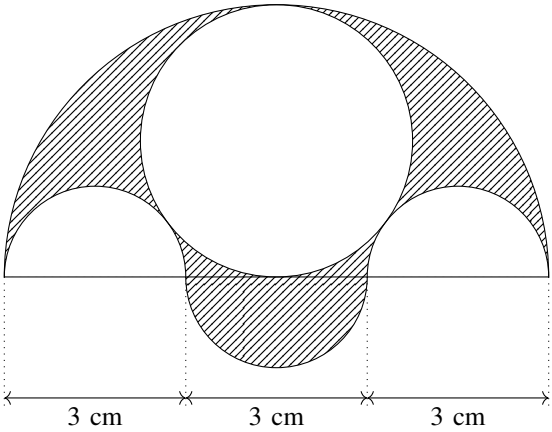


Fig. 1.1.7.1

Solution: In Fig. 1.1.7.1, the radii are

$$R = \frac{9}{2} \text{ cm (large semicircle)}, \quad r = \frac{3}{2} \text{ cm (small semicircles)}, \quad \rho = \frac{9}{4} \text{ cm (circle)} \tag{1.1.7.1}$$

The shaded region is the large semicircle, excluding the circle and the two upper small semicircles, and including the lower small semicircle. Using (1.1.8.1), the area is

$$A = \frac{1}{2}\pi R^2 - \pi\rho^2 - 2\left(\frac{1}{2}\pi r^2\right) + \frac{1}{2}\pi r^2 \tag{1.1.7.2}$$

$$= \frac{1}{2}\pi R^2 - \pi\rho^2 - \frac{1}{2}\pi r^2 \tag{1.1.7.3}$$

$$= \pi\left(\frac{81}{8} - \frac{81}{16} - \frac{9}{8}\right) = \frac{63\pi}{16} \tag{1.1.7.4}$$

Substituting $\pi = \frac{22}{7}$,

$$A = \frac{63}{16} \times \frac{22}{7} = \frac{99}{8} = 12.375 \text{ cm}^2 \tag{1.1.7.5}$$

The following code generates Fig. 1.1.7.2

codes/shaded_area.py

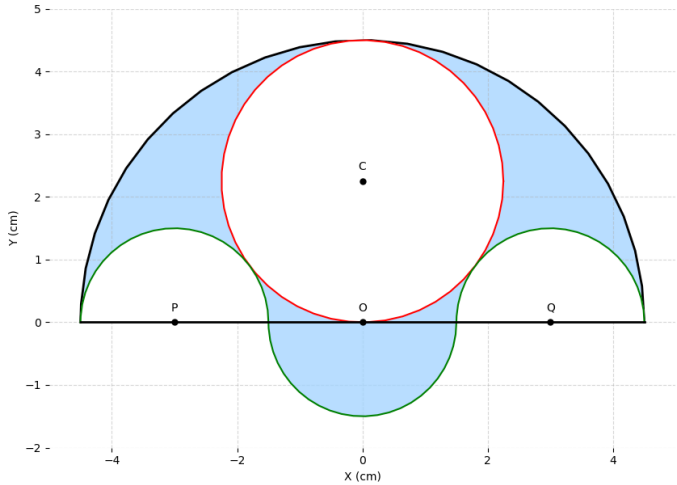


Fig. 1.1.7.2: Shaded region

1.1.8 In what ratio does the point $\left(\frac{24}{11}, y\right)$ divide the line segment joining the points $\mathbf{P}\left(2, -2\right)$ and $\mathbf{Q}\left(3, 7\right)$? Also find the value of y. (CBSE 2017)

Solution: Let the ratio be $k : 1$. From (1.1.9.1),

$$\begin{pmatrix} \frac{24}{11} \\ y \end{pmatrix} = \frac{k\begin{pmatrix} 3 \\ 7 \end{pmatrix} + \begin{pmatrix} 2 \\ -2 \end{pmatrix}}{k+1} \tag{1.1.8.1}$$

Considering the first row,

$$\frac{24}{11} = \frac{3k+2}{k+1} \tag{1.1.8.2}$$

$$\implies 24k + 24 = 33k + 22 \tag{1.1.8.3}$$

$$\implies k = \frac{2}{9} \tag{1.1.8.4}$$

Thus, the ratio is $2 : 9$. Considering the second row,

$$y = \frac{7k-2}{k+1} = \frac{\frac{14}{9} - 2}{\frac{11}{9}} = -\frac{4}{11} \tag{1.1.8.5}$$

The following code verifies the solution

codes/section.c

1.1.9 Water in a canal, 5.4 m wide and 1.8 m deep, is flowing with a speed of 25 km/hour. How much area can it irrigate in 40 minutes, if 10 cm of standing water is required for irrigation ? (CBSE 2017)

Solution: The cross-sectional area, speed and time are

$$A = 5.4 \times 1.8 \text{ m}^2, \quad v = 25000 \text{ m/h}, \quad t = \frac{40}{60} = \frac{2}{3} \text{ h} \tag{1.1.9.1}$$

From (1.1.13.1), the volume of water that flows in 40 minutes is

$$V = 5.4 \times 1.8 \times 25000 \times \frac{2}{3} = 162000 \text{ m}^3 \tag{1.1.9.2}$$

If this water stands at a depth of 10 cm $= 0.1$ m over an area S,

$$S \times 0.1 = 162000 \tag{1.1.9.3}$$

$$\implies S = 1620000 \text{ m}^2 = 162 \text{ hectares} \tag{1.1.9.4}$$

1.1.10 In the given figure, two concentric circles with centre O have radii 21 cm and 42 cm. If $\angle AOB = 60°$, find the area of the shaded region. [Use $\pi = \frac{22}{7}$] (CBSE 2017)

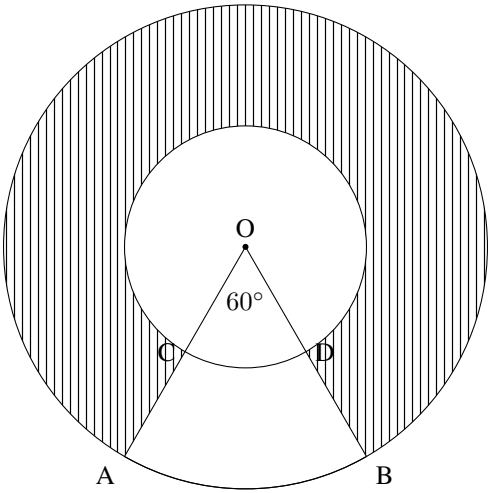


Fig. 1.1.10.1

Solution: In Fig. 1.1.10.1, the radii are $R = 42$ cm and $r = 21$ cm. The shaded region is the ring between the circles, excluding the part of the ring inside the $60°$ angle. From (1.1.15.1),

$$A = \pi\left(R^2 - r^2\right) - \frac{60°}{360°}\pi\left(R^2 - r^2\right) \tag{1.1.10.1}$$

$$= \frac{5}{6}\pi\left(42^2 - 21^2\right) = \frac{5}{6} \times \frac{22}{7} \times 1323 \tag{1.1.10.2}$$

yielding

$$A = 3465 \text{ cm}^2 \tag{1.1.10.3}$$

The following code generates Fig. 1.1.10.2

codes/sector_area.py

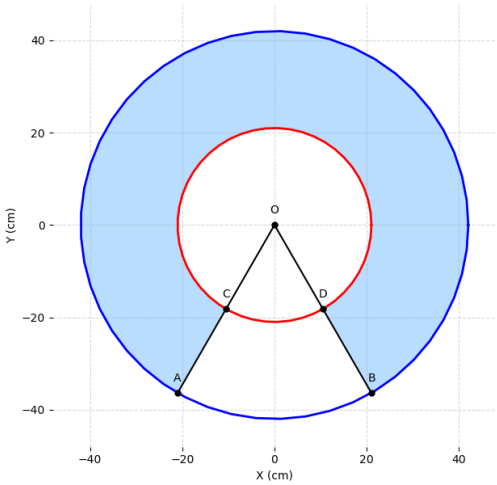


Fig. 1.1.10.2: Annular sector

1.1.11 The dimensions of a solid iron cuboid are $4.4\text{ m} \times 2.6\text{ m} \times 1.0\text{ m}$. It is melted and recast into a hollow cylindrical pipe of 30 cm inner radius and thickness 5 cm. Find the length of the pipe. (CBSE 2017)

Solution: The volume of the cuboid is

$$V = 4.4 \times 2.6 \times 1.0 = 11.44\text{ m}^3 \tag{1.1.11.1}$$

The inner and outer radii of the pipe are $r = 0.3$ m and $R = 0.3 + 0.05 = 0.35$ m. Let h be the length of the pipe. Equating the volumes using (1.1.14.1),

$$\frac{22}{7}\left(0.35^2 - 0.3^2\right)h = 11.44 \tag{1.1.11.2}$$

$$\implies \frac{22}{7} \times 0.0325\,h = 11.44 \tag{1.1.11.3}$$

$$\therefore h = 112\text{ m} \tag{1.1.11.4}$$

The following code verifies the solution

codes/cylinder.c

1.1.12 A toy is in the form of a cone of radius 3.5 cm mounted on a hemisphere of same radius on its circular face. The total height of the toy is 15.5 cm. Find the total surface area of the toy. (CBSE 2017)

Solution: The radius is $r = 3.5$ cm. The height of the cone is

$$h = 15.5 - 3.5 = 12\text{ cm} \tag{1.1.12.1}$$

From (1.1.16.1), the slant height is

$$l = \sqrt{12^2 + 3.5^2} = \sqrt{156.25} = 12.5 \text{ cm} \tag{1.1.12.2}$$

The total surface area is the sum of the curved surface areas of the cone and the hemisphere, from (1.1.16.1) and (1.1.16.2),

$$A = \pi r l + 2\pi r^2 = \pi r \left(l + 2r\right) \tag{1.1.12.3}$$

$$= \frac{22}{7} \times 3.5 \times (12.5 + 7) = 214.5 \text{ cm}^2 \tag{1.1.12.4}$$

The following code generates Fig. 1.1.12.1

codes/cone_hemisphere.py

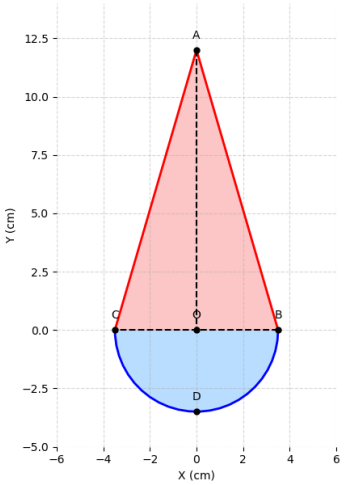


Fig. 1.1.12.1: Cone mounted on a hemisphere

1.1.13 How many terms of an A.P. $9, 17, 25, \ldots$ must be taken to give a sum of 636 ? (CBSE 2017)

Solution: Here $a = 9$ and $d = 8$. From (1.1.12.1),

$$\frac{n}{2}\left[18 + 8\left(n - 1\right)\right] = 636 \tag{1.1.13.1}$$

$$\implies 4n^2 + 5n - 636 = 0 \tag{1.1.13.2}$$

From (1.1.11.1),

$$n = \frac{-5 \pm \sqrt{25 + 10176}}{8} = \frac{-5 \pm 101}{8} \tag{1.1.13.3}$$

Since n is positive,

$$n = 12 \tag{1.1.13.4}$$

The following code verifies the solution

codes/ap_sum.c

1.1.14 If the roots of the equation $\left(a^2+b^2\right)x^2-2\left(ac+bd\right)x+\left(c^2+d^2\right)=0$ are equal, prove that $\frac{a}{b}=\frac{c}{d}$. (CBSE 2017)

Solution: From (1.1.11.2),

$$4\left(ac+bd\right)^2-4\left(a^2+b^2\right)\left(c^2+d^2\right)=0 \tag{1.1.14.1}$$

$$\implies a^2c^2+2abcd+b^2d^2=a^2c^2+a^2d^2+b^2c^2+b^2d^2 \tag{1.1.14.2}$$

$$\implies a^2d^2-2abcd+b^2c^2=0 \tag{1.1.14.3}$$

$$\implies \left(ad-bc\right)^2=0 \tag{1.1.14.4}$$

$$\implies ad=bc \tag{1.1.14.5}$$

Dividing by bd,

$$\frac{a}{b}=\frac{c}{d} \tag{1.1.14.6}$$

1.1.15 If the points $\mathbf{A}\left(k+1,2k\right)$, $\mathbf{B}\left(3k,2k+3\right)$ and $\mathbf{C}\left(5k-1,5k\right)$ are collinear, then find the value of k. (CBSE 2017)

Solution: Let

$$\mathbf{B}-\mathbf{A}=\begin{pmatrix}2k-1\\3\end{pmatrix},\quad \mathbf{C}-\mathbf{A}=\begin{pmatrix}4k-2\\3k\end{pmatrix} \tag{1.1.15.1}$$

From (1.1.10.1), the points are collinear if

$$\left|\begin{pmatrix}2k-1 & 4k-2\\3 & 3k\end{pmatrix}\right|=0 \tag{1.1.15.2}$$

$$\implies 3k\left(2k-1\right)-3\left(4k-2\right)=0 \tag{1.1.15.3}$$

$$\implies 2k^2-5k+2=0 \tag{1.1.15.4}$$

$$\implies \left(2k-1\right)\left(k-2\right)=0 \tag{1.1.15.5}$$

Hence,

$$k=\frac{1}{2}\quad \text{or}\quad k=2 \tag{1.1.15.6}$$

1.1.16 Two different dice are thrown together. Find the probability that the numbers obtained have

a) even sum, and

b) even product.

(CBSE 2017)

Solution: The total number of equally likely outcomes is $N=36$.

a) The sum is even if both numbers are even or both are odd. There are $3\times 3=9$

outcomes of each type. From (1.1.17.1),

$$\Pr\left(\text{even sum}\right) = \frac{9+9}{36} = \frac{1}{2} \tag{1.1.16.1}$$

b) The product is odd only if both numbers are odd, which gives $3 \times 3 = 9$ outcomes. Thus,

$$\Pr\left(\text{odd product}\right) = \frac{9}{36} = \frac{1}{4} \tag{1.1.16.2}$$

Using the complement in (1.1.17.1),

$$\Pr\left(\text{even product}\right) = 1 - \frac{1}{4} = \frac{3}{4} \tag{1.1.16.3}$$

The following code verifies the solution

codes/dice_prob.c

1.1.17 In the given figure, XY and $X'Y'$ are two parallel tangents to a circle with centre O and another tangent AB with point of contact C, is intersecting XY at A and $X'Y'$ at B. Prove that $\angle AOB = 90°$. (CBSE 2017)

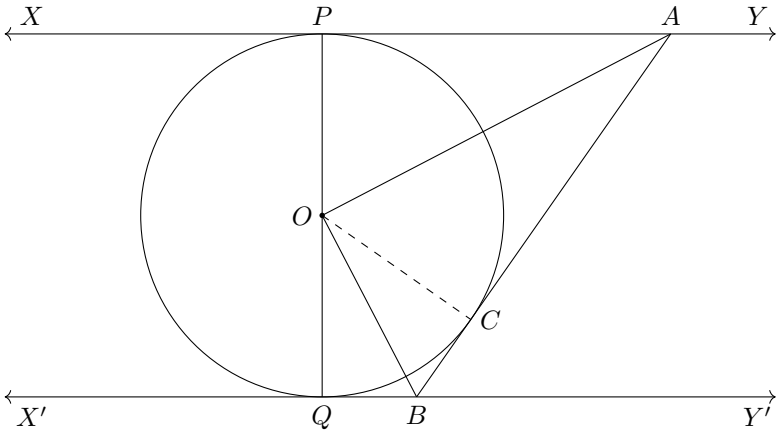


Fig. 1.1.17.1

Solution: In Fig. 1.1.17.1, let P and Q be the points of contact of XY and $X'Y'$. From (1.1.1.1), $OP \perp XY$ and $OQ \perp X'Y'$. Since $XY \parallel X'Y'$, the points P, O and Q are collinear. From (1.1.2.1),

$$AP = AC, \quad BQ = BC \tag{1.1.17.1}$$

Also $OP = OC = OQ$, and OA, OB are common. Therefore, the triangles OPA and OCA are congruent, and so are OQB and OCB,

$$\angle AOP = \angle AOC, \quad \angle BOQ = \angle BOC \tag{1.1.17.2}$$

Since POQ is a straight line,

$$\angle AOP + \angle AOC + \angle BOC + \angle BOQ = 180^\circ \tag{1.1.17.3}$$

$$\implies 2\left(\angle AOC + \angle BOC\right) = 180^\circ \tag{1.1.17.4}$$

Hence,

$$\angle AOB = \angle AOC + \angle BOC = 90^\circ \tag{1.1.17.5}$$

1.1.18 In a rain-water harvesting system, the rain-water from a roof of 22 m $\times$ 20 m drains into a cylindrical tank having diameter of base 2 m and height 3.5 m. If the tank is full, find the rainfall in cm. Write your views on water conservation. (CBSE 2017)

Solution: The radius of the tank is $r = 1$ m and its height is $h = 3.5$ m. From (1.1.18.1), the volume of the tank is

$$V = \frac{22}{7} \times 1^2 \times 3.5 = 11 \text{ m}^3 \tag{1.1.18.1}$$

Let the rainfall be x m. All the rain-water falling on the roof is collected in the tank, so

$$22 \times 20 \times x = 11 \tag{1.1.18.2}$$

$$\implies x = \frac{11}{440} = 0.025 \text{ m} \tag{1.1.18.3}$$

Hence, the rainfall is 2.5 cm. The following code verifies the solution

codes/rainfall.c

Water conservation is essential because fresh water is limited. Rain-water harvesting reduces the dependence on groundwater, recharges the water table and reduces flooding. Every individual should save water in daily life.

1.1.19 Prove that the lengths of two tangents drawn from an external point to a circle are equal. (CBSE 2017)

Solution: Let $\mathbf{P}$ be an external point of a circle with centre $\mathbf{O}$ and radius a. Let the tangents from $\mathbf{P}$ touch the circle at $\mathbf{T}_1$ and $\mathbf{T}_2$. From (1.1.1.1), the triangles $\mathbf{OT}_1\mathbf{P}$ and $\mathbf{OT}_2\mathbf{P}$ are right angled at $\mathbf{T}_1$ and $\mathbf{T}_2$. By Pythagoras' theorem,

$$\|\mathbf{P} - \mathbf{T}_1\|^2 = \|\mathbf{P} - \mathbf{O}\|^2 - \|\mathbf{T}_1 - \mathbf{O}\|^2 = \|\mathbf{P} - \mathbf{O}\|^2 - a^2 \tag{1.1.19.1}$$

$$\|\mathbf{P} - \mathbf{T}_2\|^2 = \|\mathbf{P} - \mathbf{O}\|^2 - \|\mathbf{T}_2 - \mathbf{O}\|^2 = \|\mathbf{P} - \mathbf{O}\|^2 - a^2 \tag{1.1.19.2}$$

Therefore,

$$\|\mathbf{P} - \mathbf{T}_1\| = \|\mathbf{P} - \mathbf{T}_2\| \tag{1.1.19.3}$$

1.1.20 If the ratio of the sum of the first n terms of two A.Ps is $(7n + 1) : (4n + 27)$, then find the ratio of their 9th terms. (CBSE 2017)

Solution: Let S_n and S_n' be the sums of the first n terms of the two A.Ps, and a_9,

a_9' their 9th terms. Then

$$\frac{S_n}{S_n'} = \frac{7n+1}{4n+27} \tag{1.1.20.1}$$

For $m = 9$, $2m - 1 = 17$. From (1.1.19.1),

$$\frac{a_9}{a_9'} = \frac{S_{17}/17}{S_{17}'/17} = \frac{S_{17}}{S_{17}'} = \frac{7\,(17)+1}{4\,(17)+27} = \frac{120}{95} = \frac{24}{19} \tag{1.1.20.2}$$

Thus, the ratio of the 9th terms is $24 : 19$.

1.1.21 Solve for x :

$$\frac{x-1}{2x+1} + \frac{2x+1}{x-1} = 2, \quad \text{where } x \neq -\frac{1}{2}, 1$$

(CBSE 2017)

Solution: Multiplying by $(2x+1)(x-1)$,

$$(x-1)^2 + (2x+1)^2 = 2\,(2x+1)\,(x-1) \tag{1.1.21.1}$$
$$\implies 5x^2 + 2x + 2 = 4x^2 - 2x - 2 \tag{1.1.21.2}$$
$$\implies x^2 + 4x + 4 = 0 \tag{1.1.21.3}$$
$$\implies (x+2)^2 = 0 \tag{1.1.21.4}$$

Here $q^2 - 4pr = 16 - 16 = 0$, so from (1.1.11.2) the roots are equal,

$$x = -2 \tag{1.1.21.5}$$

which satisfies $x \neq -\frac{1}{2}, 1$.

1.1.22 A takes 6 days less than B to do a work. If both A and B working together can do it in 4 days, how many days will B take to finish it ? (CBSE 2017)

Solution: Let B take x days to finish the work. Then A takes $x - 6$ days. From (1.1.20.1),

$$\frac{1}{x} + \frac{1}{x-6} = \frac{1}{4} \tag{1.1.22.1}$$
$$\implies 4\,(2x-6) = x\,(x-6) \tag{1.1.22.2}$$
$$\implies x^2 - 14x + 24 = 0 \tag{1.1.22.3}$$

From (1.1.11.1),

$$x = \frac{14 \pm \sqrt{196-96}}{2} = \frac{14 \pm 10}{2} = 12 \text{ or } 2 \tag{1.1.22.4}$$

For $x = 2$, A would take -4 days, which is not possible. Therefore, B takes 12 days. The following code verifies the solution

codes/work_time.c

1.1.23 From the top of a tower, 100 m high, a man observes two cars on the opposite sides of the tower and in same straight line with its base, with angles of depression $30°$ and $45°$. Find the distance between the cars. [Take $\sqrt{3} = 1.732$] (CBSE 2017)

Solution: Let d_1 and d_2 be the distances of the cars from the base of the tower, for the angles of depression 45° and 30° respectively. From (1.1.21.1),

$$\tan 45^\circ = \frac{100}{d_1} \implies d_1 = 100 \text{ m} \tag{1.1.23.1}$$

$$\tan 30^\circ = \frac{100}{d_2} \implies d_2 = 100\sqrt{3} \text{ m} \tag{1.1.23.2}$$

Since the cars are on opposite sides of the tower, the distance between them is

$$d_1 + d_2 = 100\left(1 + \sqrt{3}\right) = 100\left(1 + 1.732\right) = 273.2 \text{ m} \tag{1.1.23.3}$$

The following code generates Fig. 1.1.23.1

codes/tower_cars.py

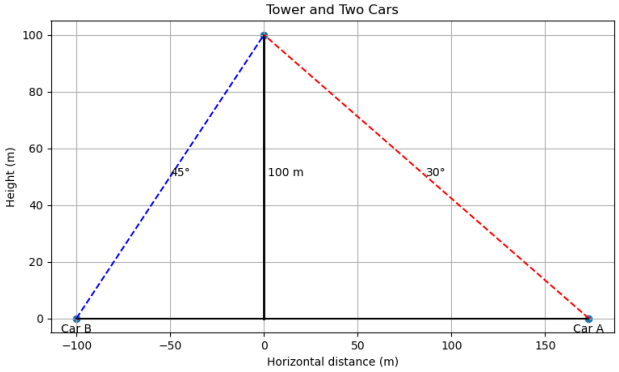


Fig. 1.1.23.1: Tower and the two cars

1.1.24 In the given figure, O is the centre of the circle with $AC = 24$ cm, $AB = 7$ cm and $\angle BOD = 90^\circ$. Find the area of the shaded region. (CBSE 2017)

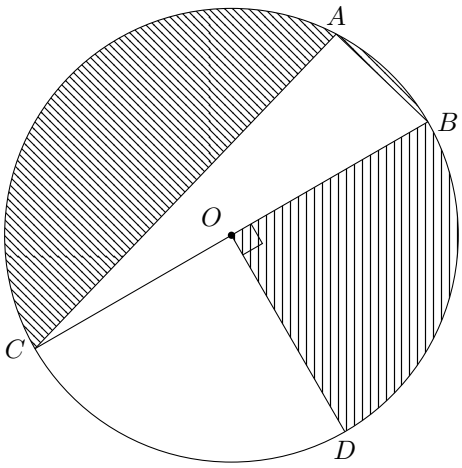


Fig. 1.1.24.1

Solution: In Fig. 1.1.24.1, B, O and C are collinear, so BC is a diameter. From (1.1.22.1),

$$\angle BAC = 90^\circ, \quad BC = \sqrt{7^2 + 24^2} = 25 \text{ cm} \tag{1.1.24.1}$$

The radius is $r = \frac{25}{2}$ cm. Since $OD \perp BC$, $\angle COD = 90^\circ$ as well. The circle consists of the triangle ABC, the two segments on AB and AC, and the two quadrants BOD and COD. The shaded region is the two segments and the quadrant BOD, so its area is

$$A = \pi r^2 - \text{ar}\,(\triangle ABC) - \text{ar}\,(\text{sector } COD) \tag{1.1.24.2}$$

$$= \pi r^2 - \frac{1}{2}\,(7)\,(24) - \frac{1}{4}\pi r^2 = \frac{3}{4}\pi r^2 - 84 \tag{1.1.24.3}$$

Substituting $\pi = \frac{22}{7}$ and $r^2 = \frac{625}{4}$,

$$A = \frac{3}{4} \times \frac{22}{7} \times \frac{625}{4} - 84 = \frac{20625}{56} - 84 = \frac{15921}{56} \approx 284.3 \text{ cm}^2 \tag{1.1.24.4}$$

The following code generates Fig. 1.1.24.2

codes/shaded_circle.py

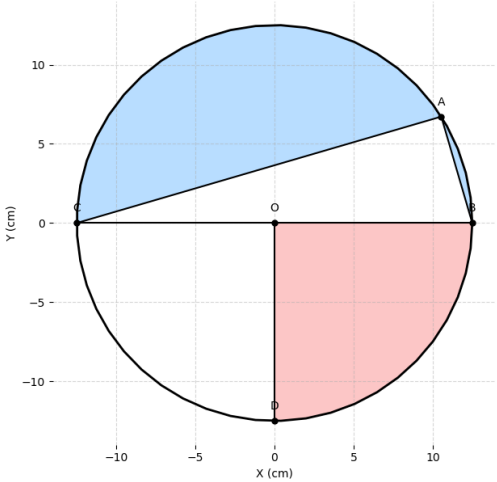


Fig. 1.1.24.2: Shaded region of the circle

B. Class 12

1) Formulae:

1.2.1 The normal form of the equation of a line is

$$\mathbf{n}^\top \mathbf{x} = c \tag{1.2.1.1}$$

The same form represents a plane when $\mathbf{x}$ and $\mathbf{n}$ are 3-dimensional vectors.

1.2.2 The distance between the parallel planes $\mathbf{n}^\top \mathbf{x} = c_1$ and $\mathbf{n}^\top \mathbf{x} = c_2$ is

$$d = \frac{|c_2 - c_1|}{\|\mathbf{n}\|} \tag{1.2.2.1}$$

Solution: The point $\mathbf{x}_1 = \frac{c_1}{\|\mathbf{n}\|^2}\mathbf{n}$ lies on the first plane, from (1.2.1.1). Moving along $\mathbf{n}$, $\mathbf{x}_1 + t\mathbf{n}$ lies on the second plane if

$$\mathbf{n}^\top (\mathbf{x}_1 + t\mathbf{n}) = c_2 \implies c_1 + t\|\mathbf{n}\|^2 = c_2 \implies t = \frac{c_2 - c_1}{\|\mathbf{n}\|^2} \tag{1.2.2.2}$$

The distance is $|t|\,\|\mathbf{n}\|$, yielding (1.2.2.1).

1.2.3 For a square matrix $\mathbf{A}$ of order n,

$$\mathbf{A}\,\mathrm{adj}\,(\mathbf{A}) = \left|\mathbf{A}\right| \mathbf{I} \tag{1.2.3.1}$$

1.2.4 (Trace of a Matrix) Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be a square matrix with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Then,

$$\sum_{i=1}^{n} \lambda_i = \sum_{i=1}^{n} a_{ii} = \mathrm{tr}\,(\mathbf{A}) \tag{1.2.4.1}$$

1.2.5 The probability mass function (PMF) of the outcome X of a fair die is

$$p_X(n) = \begin{cases} \frac{1}{6}, & 1 \le n \le 6 \\ 0, & \text{otherwise} \end{cases} \tag{1.2.5.1}$$

The probability of an event E is

$$\Pr(E) = \sum_{n\in E} p_X(n) \tag{1.2.5.2}$$

1.2.6 The events A and B are independent if and only if

$$\Pr(AB) = \Pr(A)\Pr(B) \tag{1.2.6.1}$$

1.2.7 A linear programming problem (LPP) for minimising the cost Z is

$$\text{minimise } Z = \mathbf{c}^\top \mathbf{x} \quad \text{subject to } \mathbf{M}\mathbf{x} \succeq \mathbf{b},\ \mathbf{x} \succeq \mathbf{0} \tag{1.2.7.1}$$

where the inequality $\succeq$ holds for every row (component).

1.2.8 Standard integrals,

$$\int \tan x\,dx = -\ln|\cos x| + C, \quad \int \cot x\,dx = \ln|\sin x| + C \tag{1.2.8.1}$$

Solution: Substituting $u = \cos x$, $du = -\sin x\, dx$,

$$\int \frac{\sin x}{\cos x}\, dx = -\int \frac{du}{u} = -\ln|\cos x| + C \tag{1.2.8.2}$$

Similarly, substituting $u = \sin x$, $du = \cos x\, dx$ gives the second integral.

1.2.9 For $a > 0$,

$$\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \ln\left|\frac{a+x}{a-x}\right| + C \tag{1.2.9.1}$$

Solution: Using partial fractions,

$$\frac{1}{a^2 - x^2} = \frac{1}{2a}\left[\frac{1}{a+x} + \frac{1}{a-x}\right] \tag{1.2.9.2}$$

Integrating, $\int \frac{dx}{a+x} = \ln|a+x|$ and $\int \frac{dx}{a-x} = -\ln|a-x|$, yielding (1.2.9.1).

1.2.10 (Continuity) A real-valued function $f(x)$ is continuous at $x = x_0$ if and only if

$$\lim_{x \to x_0^-} f(x) = \lim_{x \to x_0^+} f(x) = f(x_0) \tag{1.2.10.1}$$

1.2.11 (Differentiability) A function $f(x)$ is differentiable at $x = x_0$ if

$$f'\left(x_0^-\right) = \lim_{h \to 0^-} \frac{f(x_0+h) - f(x_0)}{h} = \lim_{h \to 0^+} \frac{f(x_0+h) - f(x_0)}{h} = f'\left(x_0^+\right) \tag{1.2.11.1}$$

1.2.12 If $f(x)$ is differentiable at $x = x_0$, it must necessarily be continuous at $x = x_0$.
Solution:

$$\lim_{x \to x_0} \left[f(x) - f(x_0)\right] = \lim_{x \to x_0} \left[\frac{f(x) - f(x_0)}{x - x_0}(x - x_0)\right] = f'(x_0) \cdot 0 = 0 \tag{1.2.12.1}$$

1.2.13 (Piecewise Boundary Conditions at $x = a$) Consider a function defined by differentiable branches $f_1(x)$ and $f_2(x)$,

$$f(x) = \begin{cases} f_1(x), & x \le a \\ f_2(x), & x > a \end{cases} \tag{1.2.13.1}$$

For $f(x)$ to be continuous at $x = a$,

$$\lim_{x \to a^-} f_1(x) = \lim_{x \to a^+} f_2(x) \implies f_1(a) = f_2(a) \tag{1.2.13.2}$$

For $f(x)$ to be differentiable at $x = a$, the one-sided derivatives must coincide,

$$f_1'(a) = f_2'(a) \tag{1.2.13.3}$$

1.2.14 (Newton-Raphson Method) For a real-valued differentiable function $f(x)$, the root is iteratively approximated using its first-order Taylor series expansion about the current

estimate x_n,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \tag{1.2.14.1}$$

1.2.15 (Rolle's Theorem) If $f(x)$ is continuous on $[a, b]$, differentiable on (a, b) and $f(a) = f(b)$, then there exists $c \in (a, b)$ such that

$$f'(c) = 0 \tag{1.2.15.1}$$

1.2.16 If $f'(x) > 0$ for all x in an interval, then $f(x)$ is strictly increasing on that interval.

1.2.17 If $A = xy$, where x and y vary with time t, then by the product rule

$$\frac{dA}{dt} = x\frac{dy}{dt} + y\frac{dx}{dt} \tag{1.2.17.1}$$

1.2.18 For a continuous function $f(x)$,

$$\int_0^a f(x)\, dx = \int_0^a f(a - x)\, dx \tag{1.2.18.1}$$

Solution: Substituting $x = a - t$, $dx = -dt$, the limits $x = 0, a$ become $t = a, 0$,

$$\int_0^a f(x)\, dx = -\int_a^0 f(a - t)\, dt = \int_0^a f(a - t)\, dt \tag{1.2.18.2}$$

1.2.19 Trigonometric identities,

$$\tan(\pi - x) = -\tan x, \quad \sec(\pi - x) = -\sec x \tag{1.2.19.1}$$

$$\frac{\tan x}{\sec x + \tan x} = \frac{\sin x}{1 + \sin x}, \quad \frac{1}{1 + \sin x} = \sec^2 x - \sec x \tan x \tag{1.2.19.2}$$

Solution: Multiplying the numerator and the denominator by $\cos x$ gives the first identity. For the second,

$$\frac{1}{1 + \sin x} = \frac{1 - \sin x}{1 - \sin^2 x} = \frac{1 - \sin x}{\cos^2 x} \tag{1.2.19.3}$$

which equals $\sec^2 x - \sec x \tan x$.

1.2.20 The derivatives of $\tan x$ and $\sec x$ are

$$\frac{d}{dx}\tan x = \sec^2 x, \quad \frac{d}{dx}\sec x = \sec x \tan x \tag{1.2.20.1}$$

1.2.21 The absolute value is defined as

$$|x - a| = \begin{cases} x - a, & x \geq a \\ a - x, & x < a \end{cases} \tag{1.2.21.1}$$

1.2.22 Two vectors $\mathbf{u}$ and $\mathbf{v}$ are perpendicular if and only if

$$\mathbf{u}^\top \mathbf{v} = 0 \tag{1.2.22.1}$$

1.2.23 The area of a right-angled triangle with the perpendicular sides $\mathbf{u}$ and $\mathbf{v}$ is

$$A = \frac{1}{2} \|\mathbf{u}\| \|\mathbf{v}\| \tag{1.2.23.1}$$

1.2.24 The mean and the variance of a discrete random variable X with PMF $p_X(n)$ are

$$E(X) = \sum_n n\, p_X(n), \quad \text{Var}(X) = \sum_n n^2\, p_X(n) - (E(X))^2 \tag{1.2.24.1}$$

1.2.25 (Bayes' Theorem) Let $E_1, E_2, \ldots, E_k$ be mutually exclusive events covering the sample space. For an event A,

$$\Pr(E_i|A) = \frac{\Pr(E_i)\Pr(A|E_i)}{\sum_{j=1}^{k} \Pr(E_j)\Pr(A|E_j)} \tag{1.2.25.1}$$

1.2.26 For $ab < 1$,

$$\tan^{-1} a + \tan^{-1} b = \tan^{-1}\left(\frac{a+b}{1-ab}\right) \tag{1.2.26.1}$$

1.2.27 The value of a determinant does not change if a multiple of one row is subtracted from another row. A determinant can be expanded along any row or column, using the cofactors of its entries.

1.2.28 (System of Linear Equations) The matrix equation

$$\mathbf{A}\mathbf{x} = \mathbf{b} \tag{1.2.28.1}$$

is consistent if and only if

$$\text{rank}(\mathbf{A}) = \text{rank}([\mathbf{A} \mid \mathbf{b}]) = r \tag{1.2.28.2}$$

and the solution is unique if r equals the number of unknowns.

1.2.29 The derivative of the logarithm is

$$\frac{d}{dx} \ln u = \frac{1}{u}\frac{du}{dx} \tag{1.2.29.1}$$

1.2.30 (Logarithmic Differentiation) If y is a function of x,

$$\frac{d}{dx} x^y = x^y \left(\frac{y}{x} + \ln x \frac{dy}{dx}\right), \quad \frac{d}{dx} y^x = y^x \left(\ln y + \frac{x}{y}\frac{dy}{dx}\right) \tag{1.2.30.1}$$

Solution: Taking logarithms, $\ln(x^y) = y \ln x$ and $\ln(y^x) = x \ln y$. Differentiating both sides using (1.2.29.1) and multiplying by x^y and y^x respectively yields (1.2.30.1).

1.2.31 Standard integrals,

$$\int \frac{dt}{t^2 + a^2} = \frac{1}{a}\tan^{-1}\left(\frac{t}{a}\right) + C, \quad \int \frac{f'(t)}{f(t)}\, dt = \ln|f(t)| + C \tag{1.2.31.1}$$

1.2.32 The maximum or the minimum value of a linear function Z over a feasible region that is a convex polygon occurs at a corner point of the region.

1.2.33 The points $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{D}$ are coplanar if and only if

$$\text{rank}\begin{pmatrix}\mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A} & \mathbf{D}-\mathbf{A}\end{pmatrix} < 3 \tag{1.2.33.1}$$

that is, the determinant of the matrix is zero.

1.2.34 The plane through the points $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ is

$$\mathbf{n}^\top \mathbf{x} = \mathbf{n}^\top \mathbf{A}, \quad \mathbf{n} = (\mathbf{B}-\mathbf{A}) \times (\mathbf{C}-\mathbf{A}) \tag{1.2.34.1}$$

1.2.35 The parametric form of the equation of a line is

$$\mathbf{x} = \mathbf{h} + \kappa \mathbf{m} \tag{1.2.35.1}$$

where $\mathbf{m}$ is the direction vector, and $\mathbf{m}^\top \mathbf{n} = 0$ for a normal $\mathbf{n}$ of the line.

1.2.36 (Line-Conic Intersection) The points of intersection of a line $L : \mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$ ($\kappa \in \mathbb{R}$) with a conic section $\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$ are given by

$$\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m} \tag{1.2.36.1}$$

where

$$\kappa_i = \frac{-\mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u}) \pm \sqrt{\left[\mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u})\right]^2 - g(\mathbf{h})(\mathbf{m}^\top \mathbf{V}\mathbf{m})}}{\mathbf{m}^\top \mathbf{V}\mathbf{m}}, \quad i = 1, 2 \tag{1.2.36.2}$$

with $g(\mathbf{h}) = \mathbf{h}^\top \mathbf{V}\mathbf{h} + 2\mathbf{u}^\top \mathbf{h} + f$.

1.2.37 The area enclosed between the curves $y = f(x)$ (upper) and $y = g(x)$ (lower) from $x = a$ to $x = b$ is

$$A = \int_a^b \left[f(x) - g(x)\right] dx \tag{1.2.37.1}$$

1.2.38 A function $f : X \to Y$ is bijective if it is one-one and onto,

$$f(x_1) = f(x_2) \implies x_1 = x_2, \quad \forall y \in Y,\ \exists x \in X \text{ such that } f(x) = y \tag{1.2.38.1}$$

The inverse of a bijective function is obtained by solving $y = f(x)$ for x.

1.2.39 The function $f(x) = \frac{ax+b}{cx+d}$ corresponds to the matrix $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ acting on $\begin{pmatrix} x \\ 1 \end{pmatrix}$. Composition of such functions corresponds to the matrix product, and the inverse function corresponds to $\mathbf{M}^{-1}$ when $\left|\mathbf{M}\right| \neq 0$.

1.2.40 The inverse of a square matrix $\mathbf{A}$ with $\left|\mathbf{A}\right| \neq 0$ is

$$\mathbf{A}^{-1} = \frac{\text{adj}(\mathbf{A})}{\left|\mathbf{A}\right|} \tag{1.2.40.1}$$

and the solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$ is $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.

1.2.41 The Laplace transform of a function $g(t)$ is defined as

$$g(t) \overset{\mathcal{L}}{\longleftrightarrow} G(s) = \int_0^\infty e^{-st} g(t)\, dt \tag{1.2.41.1}$$

The independent variable can have any name, such as y.

1.2.42 The Laplace transform of the derivative is

$$g'(t) \overset{\mathcal{L}}{\longleftrightarrow} sG(s) - g(0) \tag{1.2.42.1}$$

Solution: Applying integration by parts, with $u = e^{-st}$ and $dv = g'(t)\,dt$,

$$\int_0^\infty e^{-st} g'(t)\,dt = \left[e^{-st} g(t)\right]_0^\infty + s\int_0^\infty e^{-st} g(t)\,dt = -g(0) + sG(s) \tag{1.2.42.2}$$

1.2.43 Multiplication by t corresponds to differentiation with respect to s,

$$t\,g(t) \overset{\mathcal{L}}{\longleftrightarrow} -\frac{dG(s)}{ds} \tag{1.2.43.1}$$

Solution: Differentiating (1.2.41.1) with respect to s,

$$\frac{dG(s)}{ds} = \int_0^\infty (-t)\, e^{-st} g(t)\,dt \tag{1.2.43.2}$$

1.2.44 For a non-negative integer n,

$$t^n u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{n!}{s^{n+1}} \tag{1.2.44.1}$$

Solution: From $u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s}$, applying (1.2.43.1) n times gives $(-1)^n \frac{d^n}{ds^n}\frac{1}{s} = \frac{n!}{s^{n+1}}$.

1.2.45 (Shift Property) For a constant a,

$$e^{at} g(t) \overset{\mathcal{L}}{\longleftrightarrow} G(s-a) \tag{1.2.45.1}$$

1.2.46 The Laplace transforms of the cosine and sine functions are

$$\cos(at)\,u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{s}{s^2+a^2}, \quad \sin(at)\,u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{a}{s^2+a^2} \tag{1.2.46.1}$$

1.2.47 The linear system $\frac{d\mathbf{x}}{dt} = \mathbf{M}\mathbf{x}$ with $\mathbf{x}(0) = \mathbf{x}_0$ has the Laplace transform

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{M})^{-1}\mathbf{x}_0 \tag{1.2.47.1}$$

Solution: Taking the Laplace transform of each row and using (1.2.42.1),

$$s\mathbf{X}(s) - \mathbf{x}_0 = \mathbf{M}\mathbf{X}(s) \implies (s\mathbf{I} - \mathbf{M})\mathbf{X}(s) = \mathbf{x}_0 \tag{1.2.47.2}$$

1.2.48 For a 2×2 matrix $\mathbf{M}$, the poles of $(s\mathbf{I} - \mathbf{M})^{-1}$ are the eigenvalues of $\mathbf{M}$, and

$$|s\mathbf{I} - \mathbf{M}| = s^2 - \text{tr}(\mathbf{M})\,s + |\mathbf{M}| \tag{1.2.48.1}$$

The sum of the eigenvalues is $\text{tr}(\mathbf{M})$ and their product is $|\mathbf{M}|$.

2) Problems:

1.2.1 Find :

$$\int \frac{\sin^2 x - \cos^2 x}{\sin x \cos x}\,dx$$

(CBSE 2017)

Solution: Splitting the numerator,

$$\frac{\sin^2 x - \cos^2 x}{\sin x \cos x} = \frac{\sin x}{\cos x} - \frac{\cos x}{\sin x} = \tan x - \cot x \tag{1.2.1.1}$$

From (1.2.8.1),

$$\int (\tan x - \cot x)\, dx = -\ln|\cos x| - \ln|\sin x| + C \tag{1.2.1.2}$$

$$= -\ln|\sin x \cos x| + C \tag{1.2.1.3}$$

Since $\sin x \cos x = \frac{1}{2}\sin 2x$, this is also $-\ln|\sin 2x| + C'$.

1.2.2 Find the distance between the planes $2x - y + 2z = 5$ and $5x - 2.5y + 5z = 20$. (CBSE 2017)

Solution: The first plane is, from (1.2.1.1),

$$\begin{pmatrix} 2 & -1 & 2 \end{pmatrix} \mathbf{x} = 5 \tag{1.2.2.1}$$

Dividing the second equation by 2.5,

$$\begin{pmatrix} 5 & -2.5 & 5 \end{pmatrix} \mathbf{x} = 20 \implies \begin{pmatrix} 2 & -1 & 2 \end{pmatrix} \mathbf{x} = 8 \tag{1.2.2.2}$$

The planes are parallel with

$$\mathbf{n} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}, \quad c_1 = 5, \quad c_2 = 8, \quad \|\mathbf{n}\| = \sqrt{4+1+4} = 3 \tag{1.2.2.3}$$

From (1.2.2.1),

$$d = \frac{|8-5|}{3} = 1 \tag{1.2.2.4}$$

The following code verifies the solution

codes/distance.py

1.2.3 If for any 2×2 square matrix $\mathbf{A}$, $\mathbf{A}\,\text{adj}\,(\mathbf{A}) = \begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix}$, then write the value of $|\mathbf{A}|$. (CBSE 2017)

Solution: From (1.2.3.1),

$$\mathbf{A}\,\text{adj}\,(\mathbf{A}) = |\mathbf{A}|\,\mathbf{I} = \begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix} = 8\mathbf{I} \tag{1.2.3.1}$$

Comparing the entries,

$$|\mathbf{A}| = 8 \tag{1.2.3.2}$$

As a check, taking the trace of both sides using (1.2.4.1),

$$\text{tr}\left(|\mathbf{A}|\,\mathbf{I}\right) = 2\,|\mathbf{A}| = \text{tr}\,(8\mathbf{I}) = 16 \implies |\mathbf{A}| = 8 \tag{1.2.3.3}$$

The following code verifies the solution

codes/determinant.py

1.2.4 A die, whose faces are marked 1, 2, 3 in red and 4, 5, 6 in green, is tossed. Let A be the event "number obtained is even" and B be the event "number obtained is red". Find if A and B are independent events. (CBSE 2017)

Solution: The events are

$$A = \{2,4,6\}, \quad B = \{1,2,3\}, \quad AB = \{2\} \tag{1.2.4.1}$$

From (1.2.5.1) and (1.2.5.2),

$$\Pr(A) = \frac{3}{6} = \frac{1}{2}, \quad \Pr(B) = \frac{3}{6} = \frac{1}{2}, \quad \Pr(AB) = \frac{1}{6} \tag{1.2.4.2}$$

From (1.2.6.1),

$$\Pr(A)\Pr(B) = \frac{1}{4} \neq \frac{1}{6} = \Pr(AB) \tag{1.2.4.3}$$

Therefore, A and B are not independent events.
The following code verifies the solution

codes/dice.c

1.2.5 Two tailors, A and B, earn ₹ 300 and ₹ 400 per day respectively. A can stitch 6 shirts and 4 pairs of trousers while B can stitch 10 shirts and 4 pairs of trousers per day. To find how many days should each of them work and if it is desired to produce at least 60 shirts and 32 pairs of trousers at a minimum labour cost, formulate this as an LPP. (CBSE 2017)

Solution: Let A work for x days and B work for y days. The labour cost is

$$Z = 300x + 400y \tag{1.2.5.1}$$

The number of shirts and trousers stitched should be at least 60 and 32 respectively,

$$6x + 10y \geq 60 \tag{1.2.5.2}$$

$$4x + 4y \geq 32 \tag{1.2.5.3}$$

and the number of days cannot be negative, $x \geq 0$, $y \geq 0$. From (1.2.7.1), the LPP is

$$\text{minimise } Z = \begin{pmatrix}300 & 400\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} \quad \text{subject to } \begin{pmatrix}6 & 10\\4 & 4\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} \succeq \begin{pmatrix}60\\32\end{pmatrix}, \ \begin{pmatrix}x\\y\end{pmatrix} \succeq \mathbf{0} \tag{1.2.5.4}$$

Equivalently,

$$\text{minimise } Z = 300x + 400y \quad \text{subject to } 3x + 5y \geq 30, \ x + y \geq 8, \ x, y \geq 0 \tag{1.2.5.5}$$

To find the number of days, the lines $3x + 5y = 30$ and $x + y = 8$ meet at the point

given by

$$\begin{pmatrix} 3 & 5 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 30 \\ 8 \end{pmatrix} \implies x = 5, \quad y = 3 \tag{1.2.5.6}$$

The feasible region is unbounded, with the corner points

$$(0, 8), \quad (5, 3), \quad (10, 0) \tag{1.2.5.7}$$

where $(0, 8)$ and $(10, 0)$ are the points where the lines $x + y = 8$ and $3x + 5y = 30$ meet the y-axis and the x-axis respectively, and each of them satisfies the other constraint. From (1.2.32), evaluating Z at the corner points,

$$Z(0, 8) = 400(8) = 3200 \tag{1.2.5.8}$$

$$Z(5, 3) = 300(5) + 400(3) = 2700 \tag{1.2.5.9}$$

$$Z(10, 0) = 300(10) = 3000 \tag{1.2.5.10}$$

Since the region is unbounded, we check that 2700 is the true minimum. For every feasible point,

$$Z = 50(3x + 5y) + 150(x + y) \geq 50(30) + 150(8) = 2700 \tag{1.2.5.11}$$

so no feasible point gives a smaller cost. Therefore, A should work for 5 days and B for 3 days, and the minimum labour cost is ₹ 2700. As a check, $6(5) + 10(3) = 60$ shirts and $4(5) + 4(3) = 32$ pairs of trousers are produced.
The following code verifies the solution

codes/tailor_cost.py

1.2.6 Find :

$$\int \frac{dx}{5 - 8x - x^2}$$

(CBSE 2017)

Solution: Completing the square,

$$5 - 8x - x^2 = 21 - \left(x^2 + 8x + 16\right) = \left(\sqrt{21}\right)^2 - (x + 4)^2 \tag{1.2.6.1}$$

Substituting $u = x + 4$, $du = dx$, and using (1.2.9.1) with $a = \sqrt{21}$,

$$\int \frac{dx}{5 - 8x - x^2} = \int \frac{du}{\left(\sqrt{21}\right)^2 - u^2} = \frac{1}{2\sqrt{21}} \ln \left| \frac{\sqrt{21} + u}{\sqrt{21} - u} \right| + C \tag{1.2.6.2}$$

yielding

$$\int \frac{dx}{5 - 8x - x^2} = \frac{1}{2\sqrt{21}} \ln \left| \frac{\sqrt{21} + x + 4}{\sqrt{21} - x - 4} \right| + C \tag{1.2.6.3}$$

1.2.7 Find the value of c in Rolle's theorem for the function $f(x) = x^3 - 3x$ in $\left[-\sqrt{3}, 0\right]$. (CBSE 2017)

Solution: The function $f(x)$ is a polynomial, so it is differentiable, and hence continuous, from (1.2.12.1). At the end points,

$$f\left(-\sqrt{3}\right) = -3\sqrt{3} + 3\sqrt{3} = 0, \quad f(0) = 0 \tag{1.2.7.1}$$

The conditions of (1.2.15.1) hold, so there exists $c \in \left(-\sqrt{3}, 0\right)$ with $f'(c) = 0$,

$$f'(x) = 3x^2 - 3 = 3(x-1)(x+1) = 0 \tag{1.2.7.2}$$

$$\implies x = \pm 1 \tag{1.2.7.3}$$

Since $-\sqrt{3} \approx -1.732$, only $x = -1$ lies in $\left(-\sqrt{3}, 0\right)$, so

$$c = -1 \tag{1.2.7.4}$$

Using (1.2.14.1) on $f'(x)$, with $f''(x) = 6x$,

$$x_{n+1} = x_n - \frac{3x_n^2 - 3}{6x_n} = \frac{x_n^2 + 1}{2x_n} \tag{1.2.7.5}$$

For $x_0 = -1.5$, the iterations are $x_1 \approx -1.0833$, $x_2 \approx -1.0032$, $x_3 \approx -1.0000$, which converge to $c = -1$. The following code verifies the solution

codes/rolle.c

1.2.8 Show that the function $f(x) = x^3 - 3x^2 + 6x - 100$ is increasing on $\mathbb{R}$. (CBSE 2017)

Solution: Differentiating,

$$f'(x) = 3x^2 - 6x + 6 = 3\left[(x-1)^2 + 1\right] \tag{1.2.8.1}$$

Since $(x-1)^2 \geq 0$ for all $x \in \mathbb{R}$,

$$f'(x) \geq 3 > 0 \tag{1.2.8.2}$$

Hence, from (1.2.16), $f(x)$ is strictly increasing on $\mathbb{R}$.
The following code generates Fig. 1.2.8.1

codes/Increasing.py

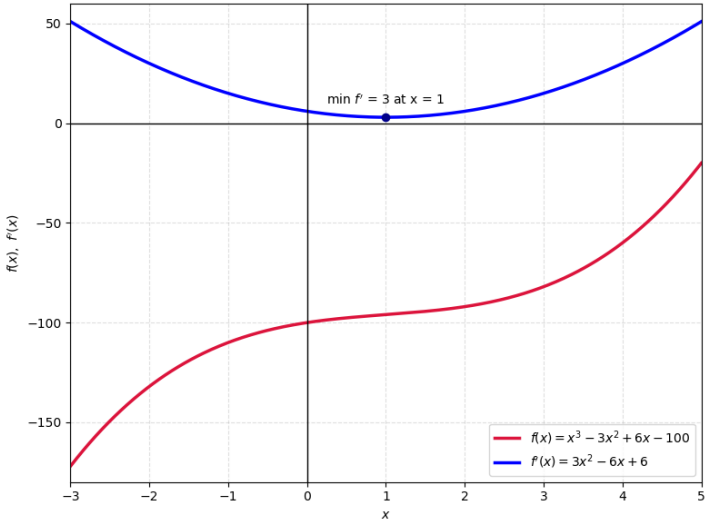


Fig. 1.2.8.1: Increasing function

1.2.9 The length x, of a rectangle is decreasing at the rate of 5 cm/minute and the width y, is increasing at the rate of 4 cm/minute. When $x = 8$ cm and $y = 6$ cm, find the rate of change of the area of the rectangle. (CBSE 2017)

Solution: The area is $A = xy$, with

$$\frac{dx}{dt} = -5 \text{ cm/min}, \quad \frac{dy}{dt} = 4 \text{ cm/min} \tag{1.2.9.1}$$

From (1.2.17.1), at $x = 8$, $y = 6$,

$$\frac{dA}{dt} = 8\,(4) + 6\,(-5) = 32 - 30 = 2 \text{ cm}^2\text{/min} \tag{1.2.9.2}$$

The area is increasing at the rate of 2 cm^2/min. The following code verifies the solution

codes/area.c

1.2.10 Evaluate :

$$\int_0^{\pi} \frac{x \tan x}{\sec x + \tan x}\, dx$$

(CBSE 2017)

Solution: Let

$$I = \int_0^{\pi} \frac{x \tan x}{\sec x + \tan x}\, dx \tag{1.2.10.1}$$

From (1.2.18.1) and (1.2.19.1),

$$I = \int_0^\pi \frac{(\pi - x)\,(-\tan x)}{-\sec x - \tan x}\,dx = \int_0^\pi \frac{(\pi - x)\tan x}{\sec x + \tan x}\,dx \tag{1.2.10.2}$$

Adding the two expressions of I,

$$2I = \pi \int_0^\pi \frac{\tan x}{\sec x + \tan x}\,dx = \pi \int_0^\pi \frac{\sin x}{1 + \sin x}\,dx = \pi \int_0^\pi \left[1 - \frac{1}{1 + \sin x}\right] dx \tag{1.2.10.3}$$

from (1.2.19.2). Using (1.2.19.2) and (1.2.20.1),

$$\int_0^\pi \frac{dx}{1 + \sin x} = \int_0^\pi \left(\sec^2 x - \sec x \tan x\right) dx = \left[\tan x - \sec x\right]_0^\pi \tag{1.2.10.4}$$

Here $\tan x - \sec x = -\frac{\cos x}{1+\sin x}$ is continuous on $[0, \pi]$, so

$$\left[-\frac{\cos x}{1 + \sin x}\right]_0^\pi = 1 - (-1) = 2 \tag{1.2.10.5}$$

Therefore,

$$2I = \pi\,(\pi - 2) \implies I = \frac{\pi\,(\pi - 2)}{2} \tag{1.2.10.6}$$

The following code generates Fig. 1.2.10.1

codes/Integral_xtanx.py

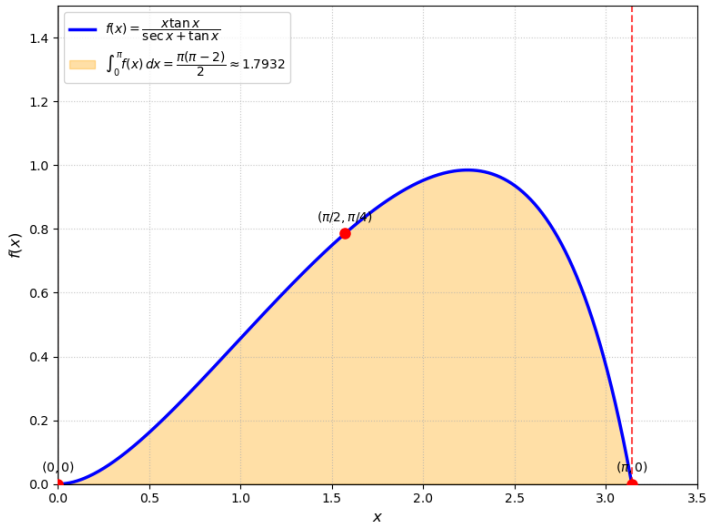


Fig. 1.2.10.1: Integral of $x \tan x$

1.2.11 Evaluate :

$$\int_1^4 \{|x-1|+|x-2|+|x-4|\}\,dx$$

(CBSE 2017)

Solution: From (1.2.21.1), for $1 \le x \le 4$, $|x-1| = x-1$ and $|x-4| = 4-x$, while $|x-2|$ changes at $x=2$. Thus the integrand is

$$g(x) = \begin{cases} (x-1)+(2-x)+(4-x) = 5-x, & 1 \le x \le 2 \\ (x-1)+(x-2)+(4-x) = x+1, & 2 < x \le 4 \end{cases} \tag{1.2.11.1}$$

At the boundary $x=2$, from (1.2.13.2),

$$f_1(2) = 5-2 = 3 = 2+1 = f_2(2) \tag{1.2.11.2}$$

so $g(x)$ is continuous. From (1.2.13.3), $f_1'(2) = -1 \neq 1 = f_2'(2)$, so $g(x)$ is not differentiable at $x=2$, but it is continuous and hence integrable. Splitting the integral at $x=2$,

$$\int_1^2 (5-x)\,dx = \left[5x - \frac{x^2}{2}\right]_1^2 = 8 - \frac{9}{2} = \frac{7}{2} \tag{1.2.11.3}$$

$$\int_2^4 (x+1)\,dx = \left[\frac{x^2}{2} + x\right]_2^4 = 12 - 4 = 8 \tag{1.2.11.4}$$

Adding,

$$\int_1^4 \{|x-1|+|x-2|+|x-4|\}\,dx = \frac{7}{2} + 8 = \frac{23}{2} \tag{1.2.11.5}$$

The following code generates Fig. 1.2.11.1

codes/Integral_modulus.py

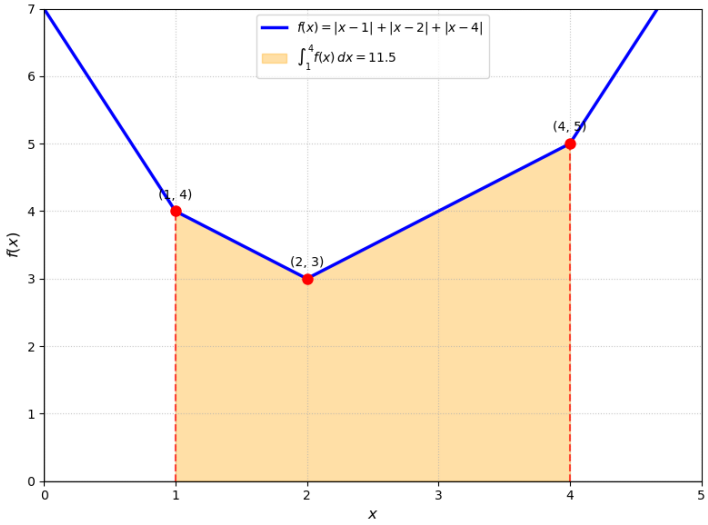


Fig. 1.2.11.1: Integral of the modulus function

1.2.12 Show that the points A, B, C with position vectors $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} - 3\hat{j} - 5\hat{k}$ and $3\hat{i} - 4\hat{j} - 4\hat{k}$ respectively, are the vertices of a right-angled triangle. Hence find the area of the triangle. (CBSE 2017)

Solution: The position vectors are

$$\mathbf{A} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 \\ -3 \\ -5 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 3 \\ -4 \\ -4 \end{pmatrix} \tag{1.2.12.1}$$

The sides are

$$\mathbf{B} - \mathbf{A} = \begin{pmatrix} -1 \\ -2 \\ -6 \end{pmatrix}, \quad \mathbf{C} - \mathbf{B} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \quad \mathbf{A} - \mathbf{C} = \begin{pmatrix} -1 \\ 3 \\ 5 \end{pmatrix} \tag{1.2.12.2}$$

From (1.2.22.1),

$$(\mathbf{B} - \mathbf{A})^{\top} (\mathbf{C} - \mathbf{B}) = -2 + 2 - 6 = -6 \neq 0 \tag{1.2.12.3}$$

$$(\mathbf{C} - \mathbf{B})^{\top} (\mathbf{A} - \mathbf{C}) = -2 - 3 + 5 = 0 \tag{1.2.12.4}$$

$$(\mathbf{A} - \mathbf{C})^{\top} (\mathbf{B} - \mathbf{A}) = 1 - 6 - 30 = -35 \neq 0 \tag{1.2.12.5}$$

Hence $CB \perp CA$, and the triangle is right angled at C. Also,

$$\|\mathbf{C} - \mathbf{B}\|^2 + \|\mathbf{A} - \mathbf{C}\|^2 = 6 + 35 = 41 = \|\mathbf{B} - \mathbf{A}\|^2 \tag{1.2.12.6}$$

From (1.2.23.1), the area is

$$A = \frac{1}{2}\sqrt{6}\sqrt{35} = \frac{\sqrt{210}}{2} \text{ sq. units} \tag{1.2.12.7}$$

The following code verifies the solution

codes/triangle.py

1.2.13 There are 4 cards numbered 1, 3, 5 and 7, one number on one card. Two cards are drawn at random without replacement. Let X denote the sum of the numbers on the two drawn cards. Find the mean and variance of X. (CBSE 2017)

Solution: There are 6 equally likely pairs, with sums

$$(1,3) \to 4, \quad (1,5) \to 6, \quad (1,7) \to 8, \quad (3,5) \to 8, \quad (3,7) \to 10, \quad (5,7) \to 12 \tag{1.2.13.1}$$

The PMF of X is

$$p_X(4) = p_X(6) = p_X(10) = p_X(12) = \frac{1}{6}, \quad p_X(8) = \frac{2}{6} = \frac{1}{3} \tag{1.2.13.2}$$

From (1.2.24.1), the mean is

$$E(X) = \frac{4+6+2(8)+10+12}{6} = \frac{48}{6} = 8 \tag{1.2.13.3}$$

and

$$\sum_n n^2 p_X(n) = \frac{16+36+2(64)+100+144}{6} = \frac{424}{6} = \frac{212}{3} \tag{1.2.13.4}$$

Therefore, the variance is

$$\text{Var}(X) = \frac{212}{3} - 64 = \frac{20}{3} \tag{1.2.13.5}$$

The following code generates Fig. 1.2.13.1

codes/PMF.py

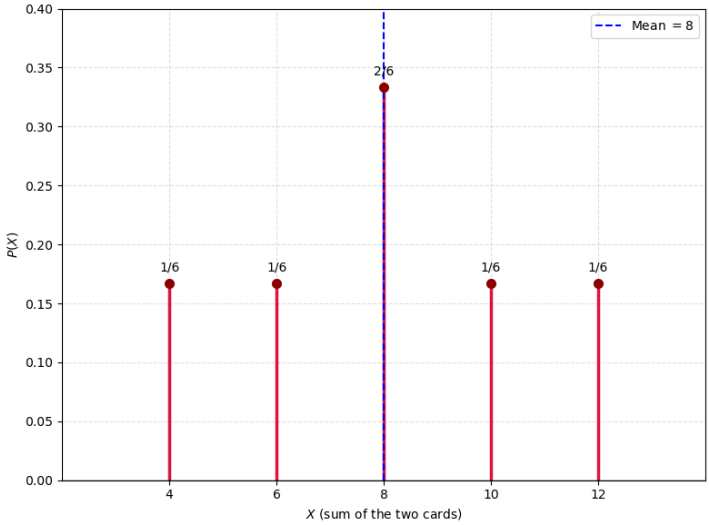


Fig. 1.2.13.1: Probability mass function

1.2.14 Of the students in a school, it is known that 30% have 100% attendance and 70% students are irregular. Previous year results report that 70% of all students who have 100% attendance attain A grade and 10% irregular students attain A grade in their annual examination. At the end of the year, one student is chosen at random from the school and he was found to have an A grade. What is the probability that the student has 100% attendance ? Is regularity required only in school ? Justify your answer. (CBSE 2017)

Solution: Let E_1 be the event that the student has 100% attendance, E_2 be the event that the student is irregular and A be the event that the student has an A grade. Then

$$\Pr(E_1) = \frac{3}{10}, \quad \Pr(E_2) = \frac{7}{10}, \quad \Pr(A|E_1) = \frac{7}{10}, \quad \Pr(A|E_2) = \frac{1}{10} \tag{1.2.14.1}$$

From (1.2.25.1),

$$\Pr(E_1|A) = \frac{\frac{3}{10}\cdot\frac{7}{10}}{\frac{3}{10}\cdot\frac{7}{10}+\frac{7}{10}\cdot\frac{1}{10}} = \frac{21}{21+7} = \frac{3}{4} \tag{1.2.14.2}$$

Regularity is not required only in school. It is needed in every area of life, such as studies, work, sports and health habits, because steady and regular effort gives better results. This is seen in the data above, where 70% of the regular students get an A grade against 10% of the irregular students. The following code verifies the solution

codes/bayes.c

1.2.15 If $\tan^{-1}\frac{x-3}{x-4} + \tan^{-1}\frac{x+3}{x+4} = \frac{\pi}{4}$, then find the value of x. (CBSE 2017)

Solution: Let

$$a = \frac{x-3}{x-4}, \quad b = \frac{x+3}{x+4} \tag{1.2.15.1}$$

Then

$$a+b = \frac{(x-3)(x+4)+(x+3)(x-4)}{x^2-16} = \frac{2x^2-24}{x^2-16} \tag{1.2.15.2}$$

$$1-ab = \frac{\left(x^2-16\right)-\left(x^2-9\right)}{x^2-16} = \frac{-7}{x^2-16} \tag{1.2.15.3}$$

Taking the tangent of both sides and using (1.2.26.1),

$$\frac{a+b}{1-ab} = \tan\frac{\pi}{4} = 1 \tag{1.2.15.4}$$

$$\implies \frac{2x^2-24}{-7} = 1 \tag{1.2.15.5}$$

$$\implies x^2 = \frac{17}{2} \tag{1.2.15.6}$$

For this value, $ab = \frac{x^2-9}{x^2-16} = \frac{1}{15} < 1$, so (1.2.26.1) holds. Hence,

$$x = \pm\sqrt{\frac{17}{2}} \tag{1.2.15.7}$$

The following code verifies the solution

codes/root.c

1.2.16 Using properties of determinants, prove that

$$\begin{vmatrix} a^2+2a & 2a+1 & 1 \\ 2a+1 & a+2 & 1 \\ 3 & 3 & 1 \end{vmatrix} = (a-1)^3$$

(CBSE 2017)

Solution: Applying $R_1 \to R_1 - R_2$ and $R_2 \to R_2 - R_3$, from (1.2.27),

$$\begin{vmatrix} a^2+2a & 2a+1 & 1 \\ 2a+1 & a+2 & 1 \\ 3 & 3 & 1 \end{vmatrix} = \begin{vmatrix} a^2-1 & a-1 & 0 \\ 2a-2 & a-1 & 0 \\ 3 & 3 & 1 \end{vmatrix} \tag{1.2.16.1}$$

Expanding along the third column,

$$= \begin{vmatrix} a^2-1 & a-1 \\ 2(a-1) & a-1 \end{vmatrix} = (a-1)(a-1)\begin{vmatrix} a+1 & 1 \\ 2 & 1 \end{vmatrix} \tag{1.2.16.2}$$

$$= (a-1)^2(a+1-2) = (a-1)^3 \tag{1.2.16.3}$$

The following code verifies the solution

codes/detcheck.py

1.2.17 Find matrix $\mathbf{A}$ such that

$$\begin{pmatrix} 2 & -1 \\ 1 & 0 \\ -3 & 4 \end{pmatrix} \mathbf{A} = \begin{pmatrix} -1 & -8 \\ 1 & -2 \\ 9 & 22 \end{pmatrix}$$

(CBSE 2017)

Solution: Let the equation be $\mathbf{BA} = \mathbf{C}$. The columns of $\mathbf{A}$ satisfy $\mathbf{Ba}_j = \mathbf{c}_j$ for $j = 1, 2$, and $\mathbf{A}$ is a 2×2 matrix. Row reducing the augmented matrix,

$$[\mathbf{B} \mid \mathbf{C}] = \begin{pmatrix} 2 & -1 & -1 & -8 \\ 1 & 0 & 1 & -2 \\ -3 & 4 & 9 & 22 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 0 & 1 & -2 \\ 2 & -1 & -1 & -8 \\ -3 & 4 & 9 & 22 \end{pmatrix} \tag{1.2.17.1}$$

$$\xrightarrow[R_3 \to R_3 + 3R_1]{R_2 \to R_2 - 2R_1} \begin{pmatrix} 1 & 0 & 1 & -2 \\ 0 & -1 & -3 & -4 \\ 0 & 4 & 12 & 16 \end{pmatrix} \xrightarrow[R_2 \to -R_2]{R_3 \to R_3 + 4R_2} \begin{pmatrix} 1 & 0 & 1 & -2 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 0 & 0 \end{pmatrix} \tag{1.2.17.2}$$

Here $\text{rank}\,(\mathbf{B}) = \text{rank}\,([\mathbf{B} \mid \mathbf{C}]) = 2$, which is the number of unknowns in each column. From (1.2.28.2), the solution exists and is unique, with

$$\mathbf{a}_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \quad \mathbf{a}_2 = \begin{pmatrix} -2 \\ 4 \end{pmatrix} \tag{1.2.17.3}$$

Hence,

$$\mathbf{A} = \begin{pmatrix} 1 & -2 \\ 3 & 4 \end{pmatrix} \tag{1.2.17.4}$$

The following code verifies the solution

codes/matrix.py

1.2.18 If $x^y + y^x = a^b$, then find $\frac{dy}{dx}$. (CBSE 2017)

Solution: Differentiating both sides with respect to x, and using (1.2.30.1),

$$x^y \left(\frac{y}{x} + \ln x \, \frac{dy}{dx} \right) + y^x \left(\ln y + \frac{x}{y} \frac{dy}{dx} \right) = 0 \tag{1.2.18.1}$$

since a^b is a constant. Collecting the terms in $\frac{dy}{dx}$,

$$\frac{dy}{dx} \left(x^y \ln x + x\, y^{x-1} \right) = - \left(y\, x^{y-1} + y^x \ln y \right) \tag{1.2.18.2}$$

Therefore,

$$\frac{dy}{dx} = - \frac{y\, x^{y-1} + y^x \ln y}{x^y \ln x + x\, y^{x-1}} \tag{1.2.18.3}$$

1.2.19 Find :

$$\int \frac{\sin\theta \, d\theta}{(4 + \cos^2\theta)\left(2 - \sin^2\theta\right)}$$

(CBSE 2017)

Solution: Substituting $t = \cos\theta$, $dt = -\sin\theta\, d\theta$ and $2 - \sin^2\theta = 1 + t^2$,

$$\int \frac{\sin\theta\, d\theta}{(4+\cos^2\theta)\left(2-\sin^2\theta\right)} = -\int \frac{dt}{(4+t^2)(1+t^2)} \tag{1.2.19.1}$$

Using partial fractions,

$$\frac{1}{(4+t^2)(1+t^2)} = \frac{1}{3}\left[\frac{1}{1+t^2} - \frac{1}{4+t^2}\right] \tag{1.2.19.2}$$

From (1.2.31.1),

$$-\frac{1}{3}\left[\tan^{-1} t - \frac{1}{2}\tan^{-1}\frac{t}{2}\right] + C \tag{1.2.19.3}$$

Substituting $t = \cos\theta$,

$$\int \frac{\sin\theta\, d\theta}{(4+\cos^2\theta)\left(2-\sin^2\theta\right)} = -\frac{1}{3}\tan^{-1}(\cos\theta) + \frac{1}{6}\tan^{-1}\left(\frac{\cos\theta}{2}\right) + C \tag{1.2.19.4}$$

1.2.20 Solve the following linear programming problem graphically :
Maximise $Z = 34x + 45y$ under the following constraints

$$x + y \le 300, \quad 2x + 3y \le 70, \quad x \ge 0,\ y \ge 0$$

(CBSE 2017)

Solution: The line $2x+3y = 70$ meets the axes at $(35, 0)$ and $\left(0, \frac{70}{3}\right)$. Since $x \le 35$ and $y \le \frac{70}{3}$ in this region, $x+y \le 300$ always holds, so that constraint is redundant. The feasible region is the triangle with the corner points $(0, 0)$, $(35, 0)$ and $\left(0, \frac{70}{3}\right)$. From (1.2.32), evaluating Z at the corner points,

$$Z(0,0) = 0 \tag{1.2.20.1}$$

$$Z(35,0) = 34(35) = 1190 \tag{1.2.20.2}$$

$$Z\left(0, \tfrac{70}{3}\right) = 45\left(\tfrac{70}{3}\right) = 1050 \tag{1.2.20.3}$$

Therefore, the maximum value of Z is 1190, at $x = 35$, $y = 0$.
The following code generates Fig. 1.2.20.1

codes/LPP.py

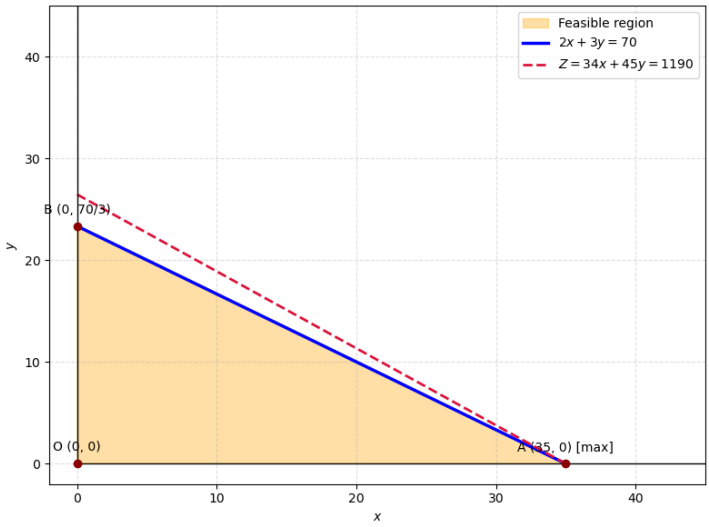


Fig. 1.2.20.1: Linear programming problem

1.2.21 Find the value of x such that the points $A\left(3,2,1\right)$, $B\left(4,x,5\right)$, $C\left(4,2,-2\right)$ and $D\left(6,5,-1\right)$ are coplanar. (CBSE 2017)

Solution: The vectors from $\mathbf{A}$ are

$$\mathbf{B}-\mathbf{A}=\begin{pmatrix}1\\x-2\\4\end{pmatrix},\quad \mathbf{C}-\mathbf{A}=\begin{pmatrix}1\\0\\-3\end{pmatrix},\quad \mathbf{D}-\mathbf{A}=\begin{pmatrix}3\\3\\-2\end{pmatrix} \tag{1.2.21.1}$$

From (1.2.33.1), the points are coplanar if

$$\begin{vmatrix}1 & x-2 & 4\\1 & 0 & -3\\3 & 3 & -2\end{vmatrix}=0 \tag{1.2.21.2}$$

Expanding along the first row,

$$1\left(0+9\right)-\left(x-2\right)\left(-2+9\right)+4\left(3-0\right)=0 \tag{1.2.21.3}$$

$$\implies 9-7\left(x-2\right)+12=0 \tag{1.2.21.4}$$

$$\implies 35-7x=0 \tag{1.2.21.5}$$

$$\therefore x=5 \tag{1.2.21.6}$$

As a check, for $x=5$, $\left(\mathbf{B}-\mathbf{A}\right)+2\left(\mathbf{C}-\mathbf{A}\right)=\begin{pmatrix}3\\3\\-2\end{pmatrix}=\mathbf{D}-\mathbf{A}$, so the rank is less than 3.

The following code verifies the solution

codes/coplanar.py

1.2.22 Find the general solution of the differential equation

$$y\,dx - \left(x + 2y^2\right)\,dy = 0$$

(CBSE 2017)

Solution: Rewriting the equation and multiplying by $y > 0$,

$$\frac{dx}{dy} - \frac{x}{y} = 2y \implies y\frac{dx}{dy} - x = 2y^2 \tag{1.2.22.1}$$

Let $x\left(y\right) \overset{\mathcal{L}}{\longleftrightarrow} X\left(s\right)$. From (1.2.43.1) and (1.2.42.1),

$$y\frac{dx}{dy} \overset{\mathcal{L}}{\longleftrightarrow} -\frac{d}{ds}\left[sX\left(s\right) - x\left(0\right)\right] = -X\left(s\right) - sX'\left(s\right) \tag{1.2.22.2}$$

From (1.2.44.1), $2y^2 \overset{\mathcal{L}}{\longleftrightarrow} \frac{4}{s^3}$. Taking the Laplace transform of (1.2.22.1),

$$-X - sX' - X = \frac{4}{s^3} \tag{1.2.22.3}$$

$$\implies X' + \frac{2}{s}X = -\frac{4}{s^4} \tag{1.2.22.4}$$

Multiplying by s^2,

$$\frac{d}{ds}\left(s^2X\right) = -\frac{4}{s^2} \tag{1.2.22.5}$$

$$\implies s^2X = \frac{4}{s} + C \tag{1.2.22.6}$$

$$\implies X\left(s\right) = \frac{4}{s^3} + \frac{C}{s^2} \tag{1.2.22.7}$$

Taking the inverse Laplace transform using (1.2.44.1),

$$x\left(y\right) = 2y^2 + Cy \tag{1.2.22.8}$$

Note that $x\left(0\right)$ cancelled out, so C remains arbitrary. As a check, $\frac{dx}{dy} = 4y + C$ and $\frac{x}{y} + 2y = 2y + C + 2y = 4y + C$. Therefore, the general solution is

$$x = 2y^2 + Cy \tag{1.2.22.9}$$

1.2.23 Find the coordinates of the point where the line through the points $(3, -4, -5)$ and $(2, -3, 1)$, crosses the plane determined by the points $(1, 2, 3)$, $(4, 2, -3)$ and $(0, 4, 3)$. (CBSE 2017)

Solution: Let $\mathbf{A} = \begin{pmatrix}1\\2\\3\end{pmatrix}$, $\mathbf{B} = \begin{pmatrix}4\\2\\-3\end{pmatrix}$, $\mathbf{C} = \begin{pmatrix}0\\4\\3\end{pmatrix}$. Then

$$\mathbf{B} - \mathbf{A} = \begin{pmatrix}3\\0\\-6\end{pmatrix}, \quad \mathbf{C} - \mathbf{A} = \begin{pmatrix}-1\\2\\0\end{pmatrix} \tag{1.2.23.1}$$

From (1.2.34.1), a normal to the plane is

$$\mathbf{n} = \begin{pmatrix} 3 \\ 0 \\ -6 \end{pmatrix} \times \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 12 \\ 6 \\ 6 \end{pmatrix} \propto \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \tag{1.2.23.2}$$

and the plane is

$$\begin{pmatrix} 2 & 1 & 1 \end{pmatrix} \mathbf{x} = 2 + 2 + 3 = 7 \tag{1.2.23.3}$$

The line through $\mathbf{P} = \begin{pmatrix} 3 \\ -4 \\ -5 \end{pmatrix}$ and $\mathbf{Q} = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$ has the direction $\mathbf{m} = \mathbf{Q} - \mathbf{P} = \begin{pmatrix} -1 \\ 1 \\ 6 \end{pmatrix}$. From (1.2.35.1),

$$\mathbf{x} = \begin{pmatrix} 3 \\ -4 \\ -5 \end{pmatrix} + \kappa \begin{pmatrix} -1 \\ 1 \\ 6 \end{pmatrix} \tag{1.2.23.4}$$

Substituting in the equation of the plane,

$$(-3) + 5\kappa = 7 \tag{1.2.23.5}$$

$$\implies \kappa = 2 \tag{1.2.23.6}$$

Therefore, the point of intersection is

$$\mathbf{x} = \begin{pmatrix} 3 \\ -4 \\ -5 \end{pmatrix} + 2 \begin{pmatrix} -1 \\ 1 \\ 6 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 7 \end{pmatrix} \tag{1.2.23.7}$$

The following code verifies the solution

codes/lineplane.py

1.2.24 Find the particular solution of the differential equation $(x - y)\frac{dy}{dx} = (x + 2y)$, given that $y = 0$ when $x = 1$. (CBSE 2017)

Solution: The equation contains the product $y\frac{dy}{dx}$, so it cannot be transformed directly. Introducing a parameter t such that

$$\frac{dx}{dt} = x - y, \quad \frac{dy}{dt} = x + 2y \tag{1.2.24.1}$$

gives $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{x+2y}{x-y}$, which is the given equation. In matrix form,

$$\frac{d}{dt}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}, \quad \begin{pmatrix} x \\ y \end{pmatrix}_{t=0} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{1.2.24.2}$$

Let $\mathbf{M} = \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}$. Then $\operatorname{tr}(\mathbf{M}) = 3$ and $\left|\mathbf{M}\right| = 3$. From (1.2.48.1),

$$\left|s\mathbf{I} - \mathbf{M}\right| = s^2 - 3s + 3 = \left(s - \frac{3}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 \tag{1.2.24.3}$$

so the eigenvalues are $\frac{3 \pm j\sqrt{3}}{2}$. From (1.2.47.1),

$$\begin{pmatrix} X(s) \\ Y(s) \end{pmatrix} = \begin{pmatrix} s-1 & 1 \\ -1 & s-2 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{s^2 - 3s + 3} \begin{pmatrix} s-2 \\ 1 \end{pmatrix} \tag{1.2.24.4}$$

Writing $s - 2 = \left(s - \frac{3}{2}\right) - \frac{1}{2}$,

$$X(s) = \frac{s - \frac{3}{2}}{\left(s - \frac{3}{2}\right)^2 + \frac{3}{4}} - \frac{1}{\sqrt{3}} \cdot \frac{\frac{\sqrt{3}}{2}}{\left(s - \frac{3}{2}\right)^2 + \frac{3}{4}} \tag{1.2.24.5}$$

$$Y(s) = \frac{2}{\sqrt{3}} \cdot \frac{\frac{\sqrt{3}}{2}}{\left(s - \frac{3}{2}\right)^2 + \frac{3}{4}} \tag{1.2.24.6}$$

Taking the inverse Laplace transform using (1.2.45.1) and (1.2.46.1), with $\theta = \frac{\sqrt{3}}{2}t$,

$$x(t) = e^{\frac{3t}{2}} \left(\cos\theta - \frac{1}{\sqrt{3}} \sin\theta\right) \tag{1.2.24.7}$$

$$y(t) = \frac{2}{\sqrt{3}} e^{\frac{3t}{2}} \sin\theta \tag{1.2.24.8}$$

To eliminate t,

$$x + 2y = 2e^{\frac{3t}{2}} \sin\left(\theta + \frac{\pi}{6}\right), \quad \sqrt{3}x = 2e^{\frac{3t}{2}} \cos\left(\theta + \frac{\pi}{6}\right) \tag{1.2.24.9}$$

$$\implies \frac{x + 2y}{\sqrt{3}x} = \tan\left(\theta + \frac{\pi}{6}\right) \tag{1.2.24.10}$$

$$\implies \tan^{-1}\left(\frac{x + 2y}{\sqrt{3}x}\right) = \frac{\sqrt{3}}{2}t + \frac{\pi}{6} \tag{1.2.24.11}$$

Also, since $x + \frac{y}{2} = e^{\frac{3t}{2}} \cos\theta$,

$$x^2 + xy + y^2 = \left(x + \frac{y}{2}\right)^2 + \frac{3}{4}y^2 = e^{3t}\left(\cos^2\theta + \sin^2\theta\right) = e^{3t} \tag{1.2.24.12}$$

$$\implies \frac{1}{2}\ln\left(x^2 + xy + y^2\right) = \frac{3t}{2} \tag{1.2.24.13}$$

Eliminating t from the last two results, the particular solution is

$$\sqrt{3}\tan^{-1}\left(\frac{x + 2y}{\sqrt{3}x}\right) - \frac{1}{2}\ln\left(x^2 + xy + y^2\right) = \frac{\pi}{2\sqrt{3}} \tag{1.2.24.14}$$

which holds on the branch through $(1, 0)$. As a check, at $(1, 0)$ the left side is $\sqrt{3} \cdot \frac{\pi}{6} - 0 = \frac{\pi}{2\sqrt{3}}$.

The following code generates Fig. 1.2.24.1

codes/euler.py

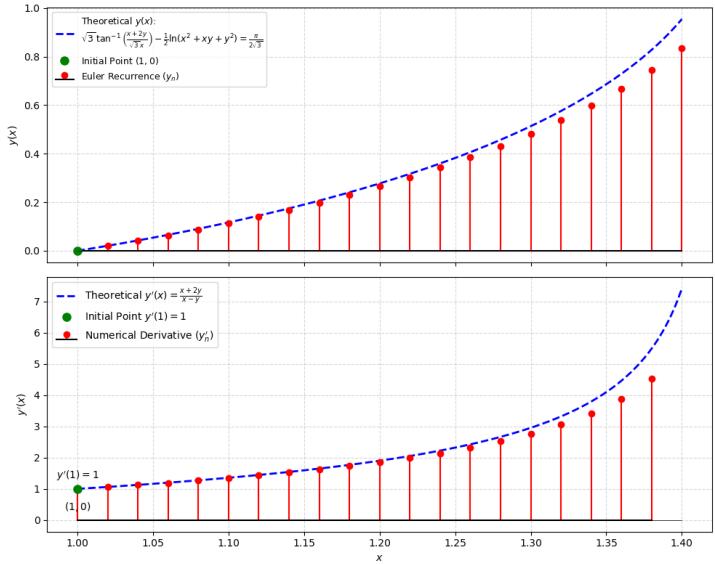


Fig. 1.2.24.1: Euler's method

1.2.25 Find the area enclosed between the parabola $4y = 3x^2$ and the straight line $3x - 2y + 12 = 0$. (CBSE 2017)

Solution: The parabola $3x^2 - 4y = 0$ is a conic with

$$\mathbf{V} = \begin{pmatrix} 3 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}, \quad f = 0 \tag{1.2.25.1}$$

The line $3x - 2y + 12 = 0$ is $\mathbf{x} = \mathbf{h} + \kappa\mathbf{m}$ with

$$\mathbf{h} = \begin{pmatrix} 0 \\ 6 \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \tag{1.2.25.2}$$

Then

$$\mathbf{Vh} + \mathbf{u} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}, \quad \mathbf{m}^\top (\mathbf{Vh} + \mathbf{u}) = -6, \quad g(\mathbf{h}) = -24, \quad \mathbf{m}^\top \mathbf{Vm} = 12 \tag{1.2.25.3}$$

From (1.2.36.1),

$$\kappa = \frac{6 \pm \sqrt{36 + 288}}{12} = \frac{6 \pm 18}{12} = 2 \text{ or } -1 \tag{1.2.25.4}$$

The points of intersection are

$$\begin{pmatrix}0\\6\end{pmatrix}+2\begin{pmatrix}2\\3\end{pmatrix}=\begin{pmatrix}4\\12\end{pmatrix},\quad \begin{pmatrix}0\\6\end{pmatrix}-\begin{pmatrix}2\\3\end{pmatrix}=\begin{pmatrix}-2\\3\end{pmatrix} \tag{1.2.25.5}$$

The line is above the parabola between $x=-2$ and $x=4$. From (1.2.37.1),

$$A=\int_{-2}^{4}\left[\frac{3x+12}{2}-\frac{3x^2}{4}\right]dx=\left[\frac{3x^2}{4}+6x-\frac{x^3}{4}\right]_{-2}^{4} \tag{1.2.25.6}$$

$$=20-(-7)=27 \text{ sq. units} \tag{1.2.25.7}$$

The following code generates Fig. 1.2.25.1

codes/parabola_line.py

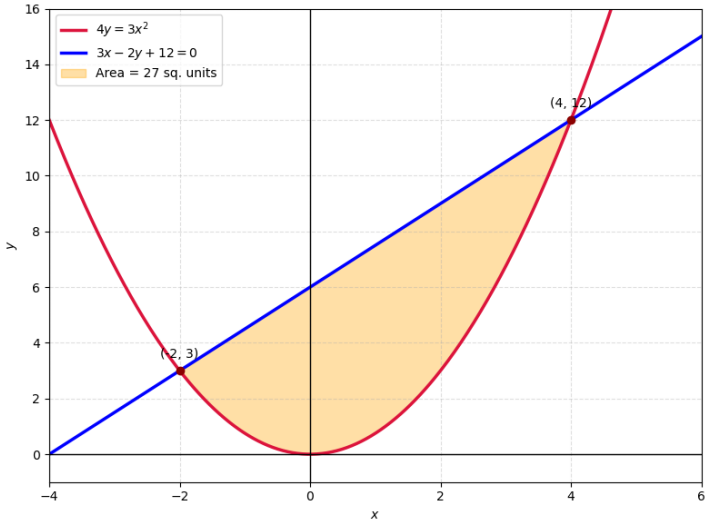


Fig. 1.2.25.1: Parabola and line

1.2.26 Consider $f:\mathbb{R}-\left\{-\frac{4}{3}\right\}\to\mathbb{R}-\left\{\frac{4}{3}\right\}$ given by $f(x)=\frac{4x+3}{3x+4}$. Show that f is bijective. Find the inverse of f and hence find $f^{-1}(0)$ and x such that $f^{-1}(x)=2$. (CBSE 2017)

Solution: From (1.2.38.1), for one-one,

$$f(x_1)=f(x_2)\implies(4x_1+3)(3x_2+4)=(4x_2+3)(3x_1+4) \tag{1.2.26.1}$$

$$\implies 7x_1=7x_2\implies x_1=x_2 \tag{1.2.26.2}$$

For onto, let $y=\frac{4x+3}{3x+4}$ with $y\neq\frac{4}{3}$. Solving for x,

$$x=\frac{3-4y}{3y-4} \tag{1.2.26.3}$$

which is defined for every $y \neq \frac{4}{3}$, and $x \neq -\frac{4}{3}$, since $x = -\frac{4}{3}$ would give $-9 = -16$. So f is onto and hence bijective. The inverse is

$$f^{-1}(x) = \frac{3-4x}{3x-4} \tag{1.2.26.4}$$

From (1.2.39), $\mathbf{M} = \begin{pmatrix} 4 & 3 \\ 3 & 4 \end{pmatrix}$ with $|\mathbf{M}| = 7 \neq 0$, and $\mathbf{M}^{-1} = \frac{1}{7}\begin{pmatrix} 4 & -3 \\ -3 & 4 \end{pmatrix}$ gives the same inverse $\frac{4x-3}{-3x+4}$. Then

$$f^{-1}(0) = \frac{3}{-4} = -\frac{3}{4} \tag{1.2.26.5}$$

and

$$f^{-1}(x) = 2 \implies 3 - 4x = 6x - 8 \implies x = \frac{11}{10} \tag{1.2.26.6}$$

1.2.27 Let $A = \mathbb{Q} \times \mathbb{Q}$ and let $*$ be a binary operation on A defined by $(a,b) * (c,d) = (ac, b + ad)$ for $(a,b), (c,d) \in A$. Determine, whether $*$ is commutative and associative. Then, with respect to $*$ on A

a) Find the identity element in A.
b) Find the invertible elements of A.

(CBSE 2017)

Solution: Writing (a,b) as the matrix $\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}$, the product

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}\begin{pmatrix} c & d \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} ac & ad+b \\ 0 & 1 \end{pmatrix} \tag{1.2.27.1}$$

is exactly $(a,b) * (c,d)$.

a) **Commutative:** $(a,b) * (c,d) = (ac, b + ad)$ whereas $(c,d) * (a,b) = (ac, d + cb)$. For $(1,2) * (3,4) = (3,6)$ and $(3,4) * (1,2) = (3,10)$. So $*$ is not commutative.
b) **Associative:**

$$[(a,b) * (c,d)] * (e,f) = (ac, b + ad) * (e,f) = (ace, b + ad + acf) \tag{1.2.27.2}$$
$$(a,b) * [(c,d) * (e,f)] = (a,b) * (ce, d + cf) = (ace, b + ad + acf) \tag{1.2.27.3}$$

The two are equal, so $*$ is associative.
c) **Identity:** Let (e_1, e_2) be the identity. Then $(a,b) * (e_1, e_2) = (ae_1, b + ae_2) = (a,b)$ for all a, b, giving $e_1 = 1$, $e_2 = 0$. The identity element is $(1,0)$, which also satisfies $(1,0) * (a,b) = (a,b)$.
d) **Invertible elements:** For (c,d) to be the inverse of (a,b),

$$(ac, b + ad) = (1,0) \implies c = \frac{1}{a}, \quad d = -\frac{b}{a} \tag{1.2.27.4}$$

which requires $a \neq 0$. So the invertible elements are (a,b) with $a \neq 0$, and the inverse is $\left(\frac{1}{a}, -\frac{b}{a}\right)$. In matrix form, this is $\left|\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}\right| = a \neq 0$.

1.2.28 If $\mathbf{A} = \begin{pmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{pmatrix}$, find $\mathbf{A}^{-1}$. Hence using $\mathbf{A}^{-1}$ solve the system of equations $2x - 3y + 5z = 11$, $3x + 2y - 4z = -5$, $x + y - 2z = -3$. (CBSE 2017)

Solution: The determinant of $\mathbf{A}$ is

$$\left|\mathbf{A}\right| = 2\left(-4+4\right) + 3\left(-6+4\right) + 5\left(3-2\right) = 0 - 6 + 5 = -1 \neq 0 \tag{1.2.28.1}$$

The cofactors are

$$C_{11} = 0,\ C_{12} = 2,\ C_{13} = 1,\ C_{21} = -1,\ C_{22} = -9,\ C_{23} = -5,\ C_{31} = 2,\ C_{32} = 23,\ C_{3} \tag{1.2.28.2}$$

so

$$\text{adj}\left(\mathbf{A}\right) = \begin{pmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{pmatrix} \tag{1.2.28.3}$$

From (1.2.40.1),

$$\mathbf{A}^{-1} = \frac{1}{-1}\begin{pmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{pmatrix} = \begin{pmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{pmatrix} \tag{1.2.28.4}$$

The system is $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 11 \\ -5 \\ -3 \end{pmatrix}$. Since $\text{rank}\left(\mathbf{A}\right) = \text{rank}\left(\left[\mathbf{A} \mid \mathbf{b}\right]\right) = 3$, from (1.2.28.2) the solution is unique. From (1.2.40.1),

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} = \begin{pmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{pmatrix}\begin{pmatrix} 11 \\ -5 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \tag{1.2.28.5}$$

Therefore,

$$x = 1, \quad y = 2, \quad z = 3 \tag{1.2.28.6}$$

The following code verifies the solution

codes/inverse.py